Historical river straightening's current effect on macroinvertebrate communities, river morphology, and habitat features



Euan Floyd

School of Natural and Environmental Science

Newcastle University

A thesis submitted for the degree of

Master of Research (MRes)

in Ecology

September 2024

Table of Contents

List of Tables **3**

List of Figures **3**

List of Equations **3**

List of Supplementary Material **3**

Acknowledgement **4**

Abstract **4**

Keywords 4

Introduction **5**

Methodology **7**

Study Site 7
Macroinvertebrate Sampling 10
Macroinvertebrate Identification 10
Chemical & Hydromorphological Measurements 10
MoRPh Habitat Survey 11
RICTS Model 12
Data Analysis 13
Taxonomic Analysis 13
Functional Analysis 14
Statistical Analysis 17

Results **18**

Chemical & Hydromorphological 18
Macroinvertebrates 19
Taxa Abundance 22
Biotic Indices 25
Functional Analysis 26
Fourth Corner Problem 28
MoRPh Habitat Survey 29
RICTS Model 30

Discussion **34**

Conclusion **44**

References **45**

Appendix **52**

List of Abbreviations 52
Supplementary Material 53

This thesis has been written to follow the author's guidelines and instructions for the academic literature journal: *Journal of Freshwater Ecology* – published by Taylor & Francis. This thesis also follows the assignment requirements for the NES8005 - MRes Research Project module at Newcastle University.

List of Tables

Table 1: Substrate size categories 11
Table 2: River MoRPh survey indices 12
Table 3: Macroinvertebrate functional traits 16
Table 4: Chemical and hydromorphological results 19
Table 5: Macroinvertebrate taxonomic groups 21
Table 6: Macroinvertebrate abundance SIMPER analysis 23
Table 7: Macroinvertebrate biotic indices 26
Table 8: Macroinvertebrate functional trait SIMPER analysis 27
Table 9: Fourth corner analysis 28
Table 10: River MoRPh survey indices result 30
Table 11: RICT EQIs 31
Table 12: RICT EQRs 32

List of Figures

Figure 1: Study site map 9
Figure 2: Average number of taxa 21
Figure 3: nMDS ordination of macroinvertebrate abundances 23
Figure 4: Cluster heatmap of macroinvertebrate abundances 24
Figure 5: Procrustes errors of RICT predicted indices 33
Figure 6: Re-naturalisation of the Breamish 38
Figure 7: Existing landscape and proposed landscape after restoration 43

List of Equations

Equation 1: LIFE Score 14
Equation 2: PSI Score 14

List of Supplementary Material

Appendix 1: Cumulative output for invert abundance SIMPER 53
Appendix 2: Macroinvertebrate abundance & environmental CCA 55
Appendix 3: Macroinvertebrate abundance & substrate composition CCA 56
Appendix 4: Cumulative output for invert traits SIMPER 57
Appendix 5: Macroinvertebrate abundance & MoRPh indices CCA 59

Acknowledgement

Much appreciation must go to my project supervisor, Dr Roy Sanderson, without your guidance and expertise, this dissertation would not have been possible. This appreciation is further extended to all my lecturers and peers at the School of Natural and Environmental Science at Newcastle University. Many thanks must also go to all the partners of the LIFE WADER project, in particular, Heather Harrison and Russell Barber from the Environment Agency. Finally, as always, thanks go out to all my family and friends. I see you!

Abstract

The widespread practice of river channelization (i.e., straightening) has caused long-lasting alterations to fluvial ecology, morphology, and habitat features. This study assessed the ecological impact of river straightening on a section of the River Breamish in Northumberland, UK, which was straightened in the Early 19th Century. Macroinvertebrate communities, physical habitat, and hydromorphology were examined across three areas: upstream, downstream, and within a straightened section of the river. Results indicate overall high ecological health and water quality; however, the straightened section exhibited slower flows, increased fine sediment, reduced taxonomic indices, and significant differences in the abundance of taxa and functional traits, such as a preference for 'slow/null flow' (e.g. *Haliplidae*) and 'epibenthic burrowing' (e.g. *Limoniidae*). Non-straightened sample areas were associated with higher diversity, coarser sediment, and faster flows, and thus associated rheophilic taxa, (e.g. *Hydropsychidae*, *Perlodidae*, and *Glossosmatidae*) and functional traits, like 'temporally attached' and 'cemented eggs'. These findings suggest that river straightening continues to homogenize habitat, altering flow regimes and substrate composition, resulting in macroinvertebrate compositional and functional changes for sustained durations. This study also highlights the downstream impact of straightening, and the ability of modified reaches to 're-naturalise' towards communities found at nearby unmodified sites. This research provides a crucial baseline for future monitoring as the river undergoes restoration towards the end of 2024, through the LIFE WADER project.

Keywords: River Channelization; Straightening; Restoration; Macroinvertebrates; Functional Traits; RICTS; MoRPh; LIFE WADER

Introduction

River channelization is the most dramatic and direct anthropogenic impact on the fluvial system (Brookes., 1988; Hohensinner *et al*., 2018). Particularly in the alluvial lowlands and valley floors, rivers and streams across the globe have undergone millennia of modification through river channelization practices (Gurnell & Downs., 2021; Brookes *et al*., 1983). The term 'river channelization' (hereby referred to as 'channelization') is synonymous with 'canalization' and refers to a variety of engineering practices conducted to improve river alignment, navigation, flooding, and the drainage of adjacent land for agricultural and development (Brookes., 1988). Channelization includes practices such as widening, dredging, straightening, embanking, bank stabilisation, and the clearance of riparian vegetation (Gregory., 2001; Hohensinner *et al*., 2018). These practices are carried out routinely and concurrently, often utilising several practices that require regular maintenance, especially in active reaches (Brookes *et al*., 1983).

There is a long history of channelization in the UK (Brookes., 1988). Sear *et al* (1999) estimated that over 35,500 km of main river has been channelized across the UK, with the true extent likely greater when including both main rivers and 'ordinary watercourses' (Sear *et al*., 1999, p.55). Although a full contemporary assessment of channelization in Britain is lacking, it was recently suggested that only 1% of rivers in the UK are 'free-flowing' and unimpeded by modification or artificial structures (Jones *et al*., 2019). Across Britain, channelization has occurred primarily as a means of flood control, particularly surrounding developed and agricultural land (Brookes., 1988; Hohensinner *et al*., 2018). While these modifications benefit society (Winiwarter *et al*., 2013), they profoundly reshape river hydrodynamics and habitat heterogeneity, consequently impacting river ecology and wider ecosystem functioning (Parker & Andres., 1976; Elosegi & Sabater., 2013; Frainer *et al*., 2018). Channelization simplifies natural flow regimes (Zhou & Endreny., 2020) and leads to the loss of natural habitat variability, both in-stream (Lau *et al*., 2006) and terrestrially through floodplain disconnection and riparian degradation (Steiger *et al*., 1998; Arnaud *et al*., 2019). The lack of floodplain connectivity is known to affect vital natural processes, such as sedimentation deposition and nutrient cycling (Hohensinner *et al*., 2004). Consequently, channelized reaches often exhibit decreased biodiversity in both riparian and aquatic communities (Nakano & Nakamura., 2008; Kennedy & Turner., 2011; Szarek-Gwiazda *et al*., 2022).

While channelization encompasses several engineering practices with different primary aims, this study will specifically examine the impacts of river straightening, which is defined as the physical modification of the river course to reduce its meanders (Zhou & Endreny., 2020). Typically, the flow of water in a naturally meandering channel induces erosion along the outside bank and deposition along the inside. This process creates an asymmetrical bathymetry, where water velocity is usually higher on the outside and lower on the inside (Dietrich & Smith., 1983; Bridges., 2003). In comparison, channelized reaches, particularly those straightened, commonly have uniform bathymetry absent from diverse flow regimes (Nakano & Nakamura., 2008). This results in a homogenisation of in-stream habitat and hydraulic complexity, which is widely recognized as a primary factor in ecological degradation (Kozarek *et al*., 2010; Zhou & Endreny., 2020).

Benthic macroinvertebrates, defined as any animal lacking a backbone and large enough to see with the naked eye (Isom., 1978), are pivotal to water resource science due to their renowned sensitivity to environmental change and habitat conditions (Feio *et al*., 2021). They serve as effective proxies for evaluating overall aquatic ecological health (Resh *et al*., 1995) and specific environmental conditions, such as flow velocity (Extence *et al*., 1999) and sedimentation (Extence *et al*., 2013). Their presence/absence, diversity, and abundance are utilized globally to assess the quality of water bodies (Kenney *et al*., 2009; Feio *et al*., 2021), serving an essential role in many national-level biomonitoring programs (Buss *et al*., 2015). Macroinvertebrate functional traits have also been used to understand the ecological conditions of freshwater systems (Lamouroux & Dolédec., 2004; Menezes *et al*., 2010). Although, they are generally less widely used compared to the more traditional taxonomic-based approaches (Bendary *et al*., 2023). It has been increasingly advocated that macroinvertebrate assessments should include both taxonomic and functional analysis (White *et al*., 2017) to provide a more comprehensive understanding the ecosystem health and functioning. This is particularly true for river restoration projects, as literature has typically assessed macroinvertebrate taxonomic response to restoration initiatives (Kali *et al*., 2015; Sinclair *et al*., 2023; Meek *et al*., 2023). However, a simple taxonomic increase is not always indicative of a more natural biological state; therefore, the inclusion of functional analysis could reveal a more accurate representation of the ecosystem's natural conditions/recovery and environmental context (Dolédec *et al*., 2015; White *et al*., 2017).

River restoration has become immensely topical in recent years (Smith *et al.*, 2014; Szałkiewicz *et al.*, 2018), focusing on rehabilitating degraded lotic systems through strategies such as floodplain reconnection and channel reconfiguration, to restore the lost habitat heterogeneity (Wohl *et al.*, 2015). Working in partnership with Natural England, the Tweed Forum, and Newcastle University, the Environment Agency (EA) successfully secured funding through the EU LIFE scheme (October 2021-2026). The LIFE WADER project seeks to improve the water quality and ecological condition across the Tweed catchment and Northumberland coast (EU LIFE., 2021). As a part of this project, the EA in partnership with AECOM consultancy are undertaking a river restoration initiative to 're-meander' the straightened reach of the Breamish, which is expected to begin by September 2024. A review by England (2021) highlighted the consistent lack of monitoring of UK river restoration projects, with only 21% of a compiled 2800 completed projects reporting any monitoring up to 2016. Particularly, pre-restoration monitoring is seldom conducted resulting in many projects lacking the means to effectively judge the ecological success of their initiative (Palmer *et al.*, 2005).

This study aims to (1) quantify differences in the taxonomic and functional composition of macroinvertebrate communities, river morphology, and habitat features between the straightened, upstream, and downstream reaches; (2) examine the functional and taxonomic dynamics of these communities in relation to environmental and hydromorphological conditions; (3) provide a pre-restoration baseline audit of macroinvertebrates and general habitat, to enable future post-restoration monitoring.

Methodology

Study Site

The River Breamish (hereby referred to as the 'Breamish') originates in the Cheviot Hills and flows in a slight north-westerly direction until its confluence with the internationally recognised River Tweed (Adams., 1999). The Breamish (within the 'Till Catchment', river's nomenclature changes to the 'River Till' immediately after the study site) is designated as a nationally important Site of Special Scientific Interest (SSSI) and Special Area of Conservation (SAC), because of its diverse landscapes, rich habitats, high biodiversity, and association with the River Tweed (Chessell., 2014). The Breamish is inhabited by several species of significant conservative value, such as the Atlantic salmon, *Salmo salar*, river/sea trout, *Salmo trutta*, lamprey spp., *Petromyzontidae*, as well as the Eurasian Otter, *Lutra lutra*, and vulnerable aquatic macrophytes like water crowfoot,

Ranunculus spp., and water starwort, *Callitriche* spp. This study focuses on a specific ~3km reach of the Breamish (Figure 1) which is situated near the hamlet of Old Bewick, Northumberland, in the northeast of England (55.484597, -1.903159). Under reference conditions, this section of the Breamish is characterised by a pool-riffle to wandering channel morphology that meanders through an alluvial valley floor fringed by fluvioglacial deposits to the west and Fell Sandstone escarpments to the east (Chessell., 2014).

As the result of a long history of anthropogenic modification within the catchment (Land Use., 2010), the study reach is straightened and confined between flood embankments. These modifications are suggested to have occurred between the late 18th and early 19th Century for flood control and to allow for agricultural usage of the fertile loamy soils of the floodplains (Sykes., 1933). The surrounding land use and cover is rural, mainly medium-large arable fields, with areas of semi-improved pastureland and estate woodlands/plantations (Land Use., 2010; Per Obs.). The straightening and embanking have resulted in a reach that is incised, with limited floodplain connection and low morphological diversity (Environment Agency., 2022a). The hydrology of the straightened section during study conditions was largely uniform with simple successions of shallow riffles, runs, and glides, and deeper slower areas overlain by silt (Per Obs.). Active management of the reach has largely ceased, which has led to extensive erosion, overtopping, and deposition along the flood banks (Environment Agency., 2022a). Considering the duration since channelization of this reach occurred and the lack of maintenance, sections of the straightened reach appear to be 're-naturalising' with small sharp meanders (Per Obs.).

This research was conducted at three classified sample areas (straightened, upstream, and downstream) along the ~3km reach (Figure 1). Five sample sites were selected within a 300m area in each sample area (n=15). Sites were chosen based on safety and ease of access, safe depth, and representative sites for each sample area. All sample areas were impacted by embankments, and only the straightened reach was impacted by straightening (Per Obs.) The original project planned for 2-3 sets of 15 samples, however, due to the Spring/Early Summer of 2024 being the second wettest on record for Solway, Tweed, and Northwest England (Chan *et al.*, 2024; Turner *et al.*, 2024), limited sampling opportunities were available when the river was not in spate, or

safe to access. All reported sampling occurred across three days in early June 2024, with the river in low flow conditions.

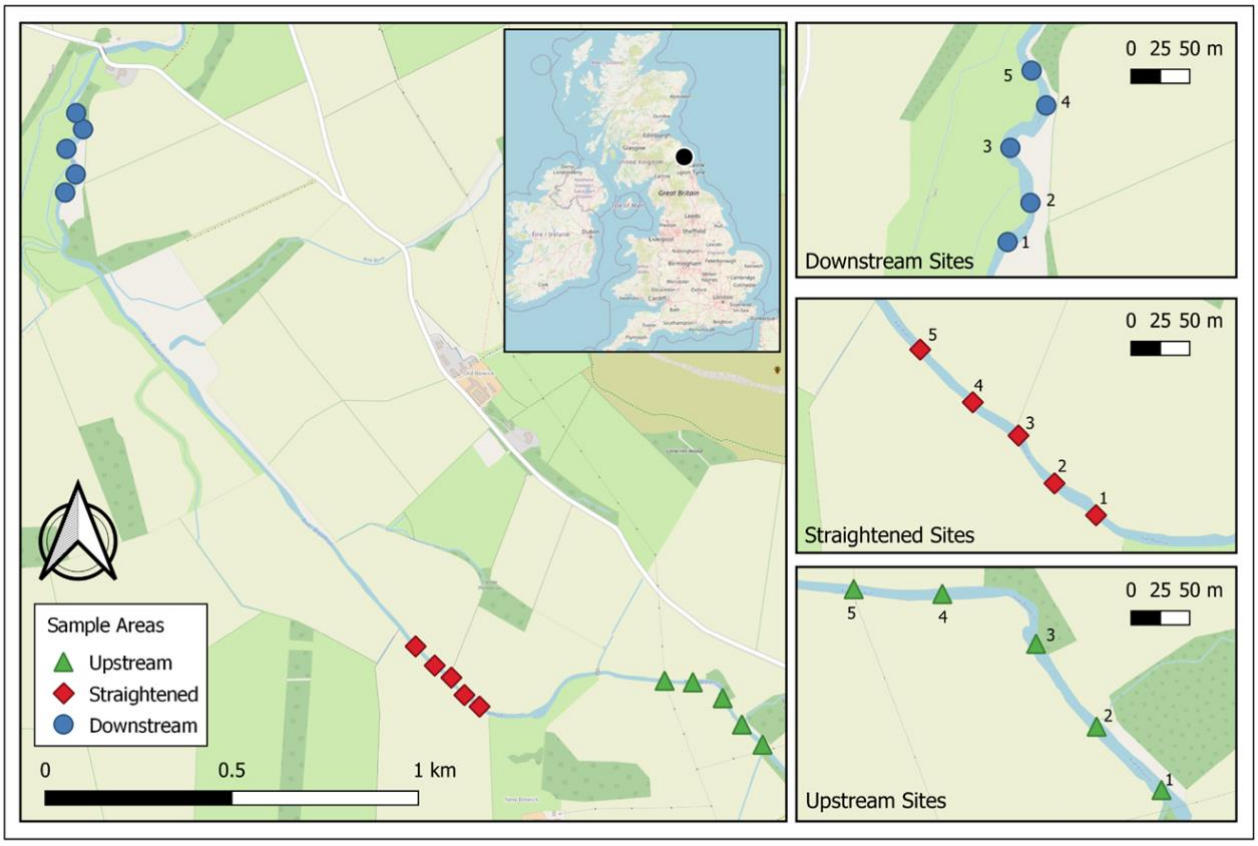


Figure 1. Study site map of the River Breamish displaying the 15 sample sites across the three sample areas.

Macroinvertebrate Sampling

15 macroinvertebrate samples were taken using a standard pond net with a multi-filament polyester oval-shaped mesh (1.4 mm ± 0.2 mm aperture), following the well-established, semi-quantitative three-minute kick sampling method (Murray-Bligh *et al.*, 2022). Each kick sample included an additional one-minute manual search of the margins and surface to include invertebrates poorly represented during kick sampling (i.e., *Gerridae*). Each three-minute sample only included time spent actively sampling, allowing for several short bursts of 15-20 seconds, which apportioned the effort across the different habitat features (e.g., riffles, pools, vegetation, sediment type) to their percentage cover within the sample site (determined visually). Following each kick sample, the contents of the net were rinsed with water to remove silt and sediment. Large debris (e.g., stones, wood, vegetation) were carefully inspected for invertebrates and discarded. The contents of the net were then decanted into 1L pails, which were kept cool and transported to the laboratory. Samples were then fixed in 70% ethanol for preservation before identification.

Macroinvertebrate Sorting and Identification

Following the Environment Agency's protocol (Murry-Bligh., 1999), small portions of the sample were placed in a white sorting tray, and all macroinvertebrates were sorted from the fine sediment/gravel and placed in a sample tube containing 70% ethanol. Using a microscope, invertebrates from each sample were identified to the lowest taxonomic level possible (mainly family), according to Pawley, Dobson, and Fletcher (2011). The total number of each taxon present in each sample was enumerated and samples were refixed in 70% ethanol and stored for quality control/audits.

Chemical and Hydromorphological Measurements

Water samples were collected from each sample site in 50 ml containers, using a 50 ml syringe and 0.2 μm nylon filter to remove bacteria, viruses, and sediment. These samples were transported back to the laboratory for chemical analysis of Nitrate (N), Phosphate (P), and Ammonium (NH_4), using standard colourimetric methods and a spectrophotometer. Water samples were filled to the top of the container, kept cool, and out of direct sunlight to limit degradation and chemical changes post-sampling (WHO, 2011). Additionally, the Myron L Ultrameter 6PIIfce was used to record water temperature, conductivity, total dissolved solids (TDS), pH, and oxidation-reduction potential (ORP) at each sample site.

Stream width and depth were determined using a 30m tape measure (width) and a one-metre ruler (depth). Stream width was quantified as the exposed water surface on the day. Stream depth was determined from an average of measurements taken from ¼, ½, and ¾ distance across the stream transect. Flow velocity was quantified using average measurements from ¼, ½, and ¾ distance across the stream transect, using a Valeport Model 801 electromagnetic flow velocity meter. The substrate composition of each sample site was estimated as percentage covers using visual interpretation from the bankside. The categories used to classify each substrate type are displayed in Table 1.

Table 1. Substrate size categories and descriptions, used to assess substrate composition.

Substrate Type	Size Range (mm)	Description
Clay	<0.06	Sticky and cohesive
Silt	<0.06	Soft, not abrasive or sticky
Sand	0.06 – 2	Abrasive, smaller than instant coffee
Gravel	2 – 16	Instant coffee to broad bean size
Pebble	16 – 64	Broad bean to half-fist size
Cobble	64 – 256	Half-fist to head size
Boulder	>256	Head size and larger

MoRPh Habitat Survey

To quantify the degree of modification, physical habitat, and hydromorphological functioning of the sample sites, The Modular River Survey (MoRPh) was conducted at each sample site. MoRPh surveys consist of many of the components utilised in the well-established River Habitat Survey (Environment Agency., 2022). However, through several modifications, MoRPh surveys allow for a more holistic approach to specific river reaches, while also being flexible, simple, and more accessible for citizen scientists (Shuker *et al*., 2017; Modular River Survey., 2019).

At each sample site, a 20m survey area was used to conduct a single MoRPh survey. Within each survey area, MoRPh survey sheets were used to record general site information, physical features, vegetation, and human modification of the bank top, bank faces, channel edges, and channel bed (Modular River Survey., 2019). The sheets use a combination of feature types, codes, and an abundance scale (A = Absent (0%), T = Trace (<5%), P = Present (5 - 33%), and E = Extensive (>33%)). Survey data were then uploaded to the MoRPh database which automatically calculates 14 indices (13 used in this study) that summarise sediment, physical habitat, vegetation, and human modification within

each sample site. As seen in Table 2, indices are split across 3 categories: channel characteristics, riparian characteristics, and human pressures & impacts.

Table 2. Index number and description of the 13 MoRPh indices utilised in this study to assess channel characteristics, riparian characteristics, and human pressures & impacts.

Index Number	Description
Channel Characteristics	
Index 1	Number of present/extensive flow types
Index 2	Highest energy present/extensive flow type (e.g. broken standing waves, upwelling, rippled)
Index 3	Number of present/extensive bed material types
Index 4	Coarsest present/extensive mineral bed material type (e.g. clay, silt, pebble, boulder)
Index 5	Average alluvial bed material size (phi units)
Index 6	Average alluvial bed material size class (e.g. clay, silt, pebble, boulder)
Index 7	Extent of superficial bed siltation
Index 8	Channel physical habitat complexity
Index 9	Number of aquatic vegetation morphotypes
Riparian Characteristics	
Index 10	Riparian physical habitat complexity
Index 11	Riparian vegetation structural complexity
Human Pressures & Impacts	
Index 12	Degree of human pressure imposed by bank top land cover
Index 13	Channel reinforcement

RICTS Model

The River Invertebrate Classification Tool (RICT) is a web tool which implements the RIVPACS IV predictive model (Murray-Bligh., 2023). RIVPACS IV is utilised to compare the observed macroinvertebrate communities to that of communities expected under 'undisturbed/minimally impacted' conditions (Gamble, 2023). As such, the presence and absence of certain species and the deviation from the expected communities could indicate changes in water quality and/or ecological health (Gamble, 2023). RICT consists of two main outputs: Prediction and Classification. Prediction is where RICT predicts the species composition and values of several indices in a 'near-natural' state.

Classification is used for the UK's official river quality assessments by comparing the predicted values with the observed values (Murray-Bligh., 2023).

Following the identification of invertebrates in the laboratory, enumerated abundances were used to calculate the Walley Hawkes Paisley Trigg (WHPT) index for each sample site. WHPT is expressed as a score, the average score per taxon (ASPT), and the number of scoring taxa (Ntaxa). WHPT index is currently used as the basis of the UK's river status classification introduced under the Water Framework Directive (Environment Agency., 2019) and is known to be sensitive to organic enrichment and toxic pollution (Murray-Bligh *et al.*, 2022). WHPT ASPT and Ntaxa, along with the recorded environmental variables were then used in RICT, following the methodology outlined in Murray-Bligh (2023). The indices predicted by RICT in this study were the Biological Monitoring Working Party (BMWP) score (Chesters., 1980), WHPT score (Environment Agency., 2019), Lotic Invertebrate Index for Flow Evaluation (LIFE) score (Extence *et al.*, 1999), and the Proportion of Sediment-Sensitive Invertebrates (PSI) score (Extence *et al.*, 2013). These index predictions were then compared with observed indices creating an Environmental Quality Index (EQI) to evaluate the degradation of macroinvertebrate communities at the sample sites relative to RICT predicted 'pristine' sample sites. RICT also classified each site, calculating Ecological Quality Ratios (EQR) and condition status classes, including the Minimum of Ntaxa and ASPT (MINTA, Murray-Bligh., 2023). Results from the RICT model in this study should be taken cautiously, as predictions and classifications were computed from only a single season and 15 samples.

Data Analysis

All data and statistical analyses were conducted in R Studios (v4.3.2) and Microsoft Excel (v16.0), or using the models implemented in RICT (Murray-Bligh., 2023), and MoRPh (Modular River Survey., 2019) survey tools.

Taxonomic Analysis

Macroinvertebrate communities from each sample site were analysed using the common taxonomic indices, Margalef richness, Pielou's evenness, and Shannon-Wiener diversity, using the 'vegan' package in R Studios (Oksanen *et al.*, 2013). BMWP scores, Ntaxa, and ASPT were also synthesised for each site. BMWP is the predecessor of the WHPT index and is a method for measuring water quality based only on the presence of

macroinvertebrates and their varying tolerances to pollutants (Fenn., 2018). Macroinvertebrate pollution tolerance scores range from 1 (very tolerant, *Oligochaeta*) to 10 (most sensitive, *Plecoptera*). BMWP scores can vary from < 10 (grossly polluted) to > 100 (unpolluted, unimpacted). The richness of *Ephemeroptera* (mayflies), *Plecoptera* (stoneflies), and *Trichoptera* (caddisflies), 'EPT scores' at each site were also calculated, due to their sensitivity to environmental changes (Ab Hamid & Rawi., 2017).

LIFE scores were quantified for each site based on recognized flow associations of different macroinvertebrate families (Extence *et al*., 1999). The methodology followed for calculating LIFE scores is outlined in Extence *et al* (1999), and the equation is:

$$\text{LIFE} = \frac{\sum fs}{n} \quad (1)$$

where $\sum fs$ is the sum of taxon flow scores for the sample, and n is the number of scoring taxa in the sample. High flow should be reflected in a high LIFE score, scores typically range between 8 for high-gradient headwaters to 6 for slower impounded rivers (Extence *et al*, 1999).

PSI scores were also calculated for each sample site. PSI is an index used to measure the impact of fine sediments on invertebrate communities (Extence *et al*., 2013). Like LIFE scores, taxa are assigned to one of four groups (A, B, C or D) depending on their tolerance to sediment levels, which are used to assign index values depending on their abundance category. PSI scores ranged between 0 (silt-free) to 100 (entirely silted). PSI is calculated using this equation (Extence *et al*., 2013):

$$\text{PSI} = \frac{\sum Scores\ for\ Sediment\ Groups\ A\ \&\ B}{\sum Scores\ for\ all\ Sediment\ Groups} \quad (2)$$

Functional Analysis

Functional traits, referred to by their 'grouping feature' and species-specific 'traits' (Schmera *et al*., 2015) were derived from a database initially developed in France, but applicable for use across all European freshwater systems (Tachet *et al*., 2010). This study used 12 'grouping features' comprising 69 'traits' (Table 3). Traits were assigned using a 'fuzzy coding' methodology (Chevene *et al*., 1994; White *et al*., 2017), wherein scores range from zero (negligible affinity) to three or five (reflecting the highest affinity per existing literature; Chevene *et al*., 1994). Affinity scores were then normalised as proportions within each grouping feature to a cumulative sum of 1 for each taxon, and a

trait-abundance matrix was then created by multiplying the taxon-grouping features by the Hellinger transformed abundances (White *et al.*, 2017). Functional trait indices were calculated using Margalef richness (trait richness), Pielou's evenness (trait evenness), and Shannon-Wiener diversity (trait diversity), in the 'vegan' package.

Table 3. Macroinvertebrate functional traits, respective codes and their modalities used within this study. (Tachet *et al.*, 2010)

Grouping Feature	Code	Traits
Maximum Potential Size	MPS1	≤ 0.25 cm
	MPS2	> 0.25-0.5 cm
	MPS3	> 0.5-1 cm
	MPS4	> 1-2 cm
	MPS5	> 2-4 cm
Life Cycle Duration	LSD1	≤ 1 year
	LSD2	> 1 year
Potential Number of Reproductive Cycles per Year	RC1	< 1
	RC2	1
	RC3	> 1
Aquatic Stages	AS1	Egg
	AS2	Larva
	AS3	Pupa
	AS4	Adult
Reproduction	R1	Ovoviviparity
	R2	Isolated eggs, free
	R3	Isolated eggs, cemented
	R4	Clutches, cemented or fixed
	R5	Clutches, free
	R6	Clutches, in vegetation
	R7	Clutches, terrestrial
	R8	Asexual reproduction
Dissemination	D1	Aquatic passive
	D2	Aquatic active
	D3	Aerial passive
	D4	Aerial active
Resistance Form	RF1	Eggs, statoblasts
	RF2	Cocoons
	RF3	Housing against desiccation
	RF4	Diapause or dormancy
	RF5	None
Locomotion and Substrate Relation	LSR1	Flier
	LSR2	Surface swimmer
	LSR3	Full water swimmer
	LSR4	Crawler
	LSR5	Burrower (epibenthic)
	LSR6	Interstitial (endobenthic)
	LSR7	Temporary attached
	LSR8	Permanently attached
Food	F1	Microorganisms
	F2	Detritus (≤ 1 mm)
	F3	Plant detritus (≥ 1 mm)
	F4	Living macrophytes
	F5	Living microphytes
	F6	Dead animals (> 1mm)
	F7	Living microinvertebrates
	F8	Living macroinvertebrates
	F9	Vertebrates
Feeding Habits	FH1	Absorber
	FH2	Deposit feeder
	FH3	Shredder
	FH4	Scraper
	FH5	Filter feeder
	FH6	Piercer (plants or animals)
	FH7	Predator (carver/engulfer/swallower)
	FH8	Parasite
Substrate (preferendum)	S1	Flags/boulders/cobbles/pebbles
	S2	Gravel
	S3	Sand
	S4	Silt
	S5	Macrophytes
	S6	Microphytes
	S7	Twigs/roots
	S8	Organic detritus/litter
	S9	Mud
Current Velocity (preferendum)	CV1	Null
	CV2	Slow (< 25 cm s^{-1})
	CV3	Medium (25-50 cm s^{-1})
	CV4	Fast (> 50 cm s^{-1})

Statistical Analysis

Data were first tested for normality and equal variance, and transformations were applied where applicable. Macroinvertebrate abundance data were transformed using Hellinger standardisation, and environmental variables were standardised to zero mean and unit standard deviation, given that they were measured across different numerical scales. Univariate and multivariate statistics were used to analyse and visualise all data.

Macroinvertebrate biotic indices were compared between sample areas using linear models and Analysis of Variance (ANOVA), in cases of statistical significance ($p < 0.05$), the post hoc Tukey HSD test was performed. Macroinvertebrate taxonomic community composition was analysed using Permutational Analysis of Variance (PERMANOVA) and Similarity Percentage Analysis (SIMPER) to investigate statistically significant differences in the taxon abundances between sample areas. Differences between the sites were visualised using non-metric multidimensional scaling (nMDS, using Bray-Curtis Similarity Matrix) and constrained ordination with the environmental variables using Canonical Correspondence Analysis (CCA). Hierarchical clustering of the taxa abundances was also conducted to produce a species abundance heatmap, in the 'tabasco' function in 'vegan' (Oksanen *et al*., 2013), which visualises the ecological dissimilarity between sites, using the Bray-Curtis Index.

Functional indices were tested using linear models and ANOVAs. The macroinvertebrate trait-abundance matrix was statistically analysed using PERMANOVA and SIMPER analysis to assess the significance and dissimilarity in the functional traits expressed between sample areas. The fourth-corner problem (Dray & Legendre., 2008) was investigated using the 'ade4' R package (Dray & Siberchicot., 2017). The fourth-corner problem reflects that matrices of samples x taxa, samples x environment, and taxa x traits can be obtained via field survey and the literature, but it is more challenging to calculate a matrix of traits x environment; the latter can provide valuable insights into the ecological function of the system (Legendre *et al*., 1997).

The MoRPh habitat survey calculated indices were tested for statistical differences between sample areas using linear models and ANOVA, in cases of significance ($p < 0.05$), the post hoc Tukey HSD test was performed. MoRPh indices were also tested against the

macroinvertebrate abundances in a CCA to assess the influence of the indices on the abundance variation across the sample areas.

The observed biotic indices were compared to the RICT-predicted biotic indices were compared in simple observed/expected ratios (EQIs). RICT calculated EQRs were analysed using linear models and ANOVAs. To compare the biotic indices predicted by RICT for the straightened sites, with differences in the input environmental variables used. A Procrustes rotation (Oksanen *et al.*, 2013) was conducted between a Principal Components Analysis (PCA) of the RICT predicted indices using observed environmental variables, and a PCA of RICT predicted indices using the upstream environmental variables, as a proxy for an unmodified condition. The alignment of these two PCAs will visualise the influence that the input environmental variables have on RICT predictions.

Results

Chemical & Hydromorphological

No detectable amount of phosphate was observed in any water sample across all sites (Table 4). There appeared to be an average accumulation of nitrate as you move downstream (Table 4). Upstream had the lowest average nitrate (2.49 ppm), followed by straightened (3.06 ppm), then downstream (4.55 ppm), yet differences were not statistically significant. Ammonium was observed to spike on average in the straightened reach (1.35 ppm), in comparison to upstream (0.05 ppm) and downstream (0.43 ppm) sites (Table 4). ANOVAs of the nitrate and ammonium measurements between sample areas revealed only the difference in nitrate to be statistically significant ($F_{2,12}$=11.95, p=0.001). The average pH was observed to be largely uniform across sites (7.6 - 7.9), displaying a slightly alkaline environment (Table 4). Both conductivity and TDS were seen to accumulate across the sample areas (Table 4), with the highest average values downstream (192.5 µS/cm, 125.3 ppm). ORP was observed to decline across the sample areas, with the lowest average value at the downstream (109.2 mV) sites, indicating low dissolved oxygen. Average water temperature appeared to fall from the upstream sites (16°C) to the straightened (15°C) and downstream (15.1°C) sites (Table 4). ANOVAs between the chemical measurements at each site showed only ORP ($F_{2,12}$=18.95, p<0.001) and temperature ($F_{2,12}$=9.25, p=0.004) to be a statistically significant difference. Post hoc Tukey HSD tests found the significant difference to be between all areas for ORP (p<0.05), particularly between up- and downstream (p=0.001). For temperature, the

difference was between the upstream and straightened (p=0.005), upstream and downstream (p=0.011).

Straightened sites were found to be on average deeper (478 mm) with a slower flow velocity (0.4 m/s), compared to upstream (353 mm, 0.7 m/s) and downstream (373 mm, 0.6 m/s). The average stream width was observed to be largely similar between sample areas, yet slightly decreasing downstream (Table 4). ANOVAs found there to be no statistically significant difference between any hydromorphological measurements at each sample area (p>0.05); however, a borderline non-significant p-value was produced for depth ($F_{2,12}$=3.67, p=0.057). The upstream site's substrate composition was on average dominated by coarse sediment (size>gravel, 87%) compared to fine sediment (size<sand, 13%). The average percentage of fine sediment increased greatly in the straightened (40%) section, and coarse sediment decreased (60%). Downstream sites displayed a similar substrate composition to the straightened reach (Table 4), although slightly increased in coarse (62%) compared to fine (38%). ANOVAs found only %pebble to differ significantly between sites ($F_{2,12}$=8.20, p=0.006), with a Tukey HSD test finding the difference to be between upstream and straightened (p=0.008), and upstream and downstream (p=0.016). A borderline nonsignificant p-value was produced for %silt ($F_{2,12}$=3.78, p=0.053).

Table 4. Averaged results of the chemical and hydromorphological analysis of the three sample areas across the Breamish. Values in bold indicate the lowest value. Highlighted values indicate the highest value.

Variable	Upstream	Straightened	Downstream
Hydromorphological			
Depth (mm)	**353.0 ± 61.9**	478.0 ± 108.3	373.3 ± 52.9
Width (m)	7.9 ± 1.4	7.8 ± 1.2	**6.9 ± 1.8**
Velocity (m/s)	0.7 ± 0.1	**0.4 ± 0.3**	0.6 ± 0.2
Coarse Sediment (%)	87 ± 7.6	**60 ± 22.1**	63 ± 14.8
Fine Sediment (%)	**13 ± 7.6**	40 ± 22.1	38 ± 14.8
Chemical			
Temperature (°C)	16.0 ± 0.6	**15.0 ± 0.1**	15.1 ± 0.3
Conductivity (µS/cm)	**177.9 ± 17.4**	181.0 ± 1.0	192.5 ± 1.1
TDS (ppm)	**115.4 ± 12.4**	117.9 ± 0.6	125.3 ± 1.3
pH	**7.6 ± 0.3**	7.9 ± 0.1	7.8 ± 0.1
ORP (mV)	157.6 ± 17.8	131.8 ± 6.4	**109.2 ± 10.2**
N (ppm)	**2.5 ± 0.5**	3.1 ± 0.6	4.5 ± 0.8
P (ppm)	0	0	0
NH_4 (ppm)	**0.05 ± 0.05**	1.3 ± 2.0	0.4 ± 0.9

Macroinvertebrates

Across the whole study, 28 macroinvertebrate families (29 including Oligochaeta (subclass)) with a total of 6,174 individuals were sampled and identified. These families/subclass were spread across 10 distinct orders and one class (Table 5). The four most abundant families were *Ephemerellidae* (spiny crawler mayflies, n=3315), *Baetidae* (small minnow mayflies, n=1323), *Elmidae* (riffle beetles, n=520), particularly *Limnius* sp. larvae (n=341), and *Limoniidae* (Limoniid crane flies, n=268). The five most abundant orders were Ephemeroptera (mayflies, n=4793), Coleoptera (beetles, n=532), Diptera (true flies, n=424), Trichoptera (caddisflies, n=226), Plecoptera (stoneflies, n=126). The five most taxa-rich orders were Trichoptera (n=8), Diptera (n=5), Ephemeroptera (n=4), followed by Coleoptera (n=3) and Plecoptera (n=3). All other orders were characterised by one taxon. Figure 2 displays the average number of observed taxa at each sample area, the downstream sites were found to have the highest average number of taxa (525.4 ± 121), followed by the upstream sites (463.4 ± 145), then the straightened sites (246 ± 187).

Table 5. Macroinvertebrates taxonomic orders and classes and their respective families and subclasses examined in the study, sampled from the river Breamish, Northumberland.

Orders/Class	**Families/Subclass**
Amphipoda	*Gammaridae*
Clitellata (class)	Oligochaeta (subclass)
Coleoptera	*Curculionidae*
	Elmidae
	Haliplidae
Cycloneritida	*Neritidae*
Diptera	*Chironomidae*
	Empididae
	Limoniidae
	Pediciidae
	Simuliidae
Ephemeroptera	*Baetidae*
	Ephemerellidae
	Ephemeridae
	Heptageniidae
Isopoda	*Asellidae*
Plecoptera	*Capniidae*
	Perlidae
	Perlodidae
Rhynchobdellida	*Glossiphoniidae*
Sphaeriida	*Sphaeriidae*
Trichoptera	*Beraidae* (excl. *Beraeodes*)
	Glossosomatidae
	Goeridae
	Hydropsychidae
	Leptoceridae
	Limnephilidae
	Psychomyiidae
	Rhyacophilidae

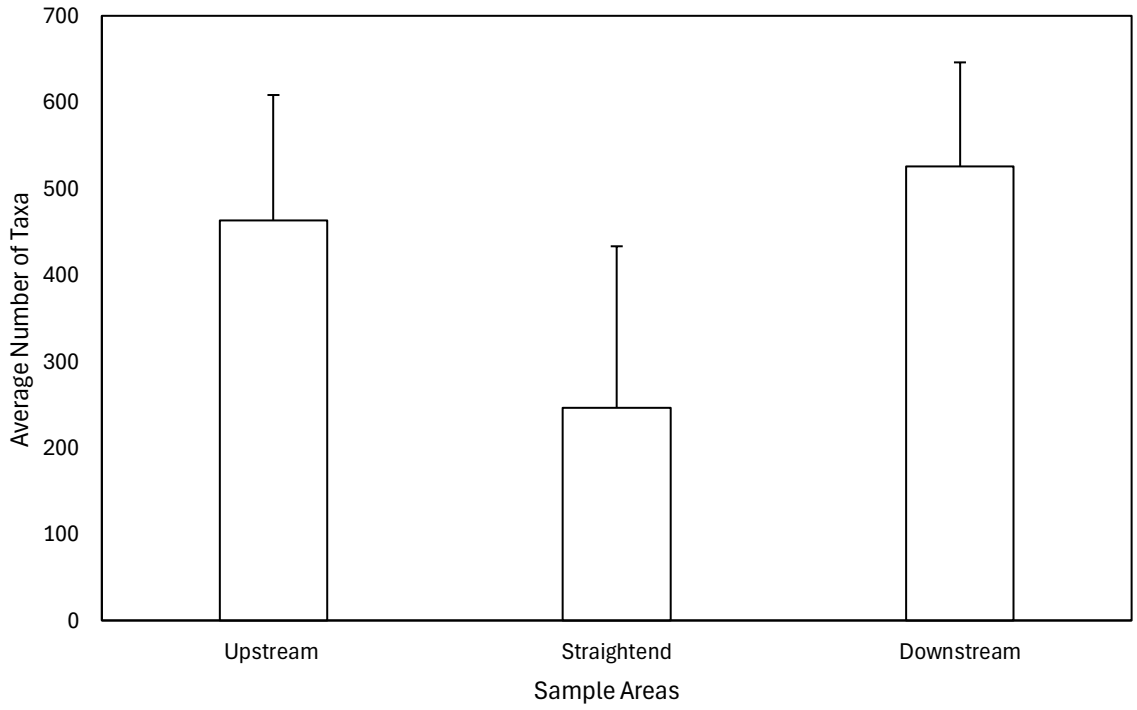


Figure 2. Average number of taxa sampled from each of the three sample areas, with standard deviation.

Taxa Abundance and Composition

The nMDS ordination for macroinvertebrate abundances at each site had a great goodness-of-fit with a stress value of 0.10. Figure 3 visualises the dissimilarity between the upstream and straightened sites and the increased similarity between upstream and downstream. Although, a slight overlap between downstream and straightened is evident. A wider variation is also seen in the taxonomic composition of the straightened sites, with two sites being more like the up-and downstream sites, while the other three sites were much more different (Figure 3). This was supported by PERMANOVA analysis of the macroinvertebrate abundances, which displayed a statistically significant difference between sample areas (pseudo-$F_{2,12}$=1.97, $p<0.05$). Yet only 24.68% of the variation in the abundances is explained by the difference between sample areas (R^2=0.2468). Based on SIMPER analysis, average dissimilarity was highest between upstream & straightened (0.0095 ± 0.0072) and downstream and straightened (0.0092 ± 0.008) sites, with the lowest dissimilarity in abundances between upstream and downstream (0.0070 ± 0.0053). Despite the limited dissimilarity observed between the sample areas, proportions of specific taxa did differ statistically as seen in Table 6 (cumulative results in Appendix 1). For instance, *Haliplidae* and *Limoniidae* were found to be significantly ($p<0.01$) more abundant in the straightened section compared to upstream and downstream. While *Hydropsychidae*, *Perlodidae*, and *Glossosmatidae* were significantly ($p<0.05$) more abundant in the upstream sites than in the straightened. In the downstream sites, *Elmidae*: *Limnius* sp. larvae were significantly more abundant than in the straightened ($p<0.05$). No statistically significant differences occurred between the upstream and downstream macroinvertebrate abundances (Table 6). Other notable differences in abundances were observed between sample areas, although not found to be statistically significant (see Appendix 1).

Table 6. Statistically significant SIMPER analysis outputs between the macroinvertebrate abundances between the 3 sampled areas. ava: average abundance in the first group. avb: average abundance in the second group. Taxa that were more abundant in the straightened section are indicated in bold. Highlighted ava & avb indicate the highest abundance in comparison to sample areas. * $p < 0.05$. ** $p < 0.001$.

Upstream vs Straightened				Straightened vs Downstream			
Species	ava (up)	avb (str)	p	Species	ava (str)	avb (ds)	p
Haliplidae	0	0.090	**	*Elmidae (Limnius)*	0.15	0.26	*
Hydropsychidae	0.14	0.06	*	***Haliplidae***	0.09	0.01	*
Limoniidea	0.18	0.27	**	***Limoniidea***	0.27	0.19	*
Perlodidae	0.05	0	*	-	-	-	-
Glossosmatidae	0.06	0.03	*	-	-	-	-
Psychomyiidae	0.02	0.01	*	-	-	-	-
Chironomidae	0	0.03	*	-	-	-	-

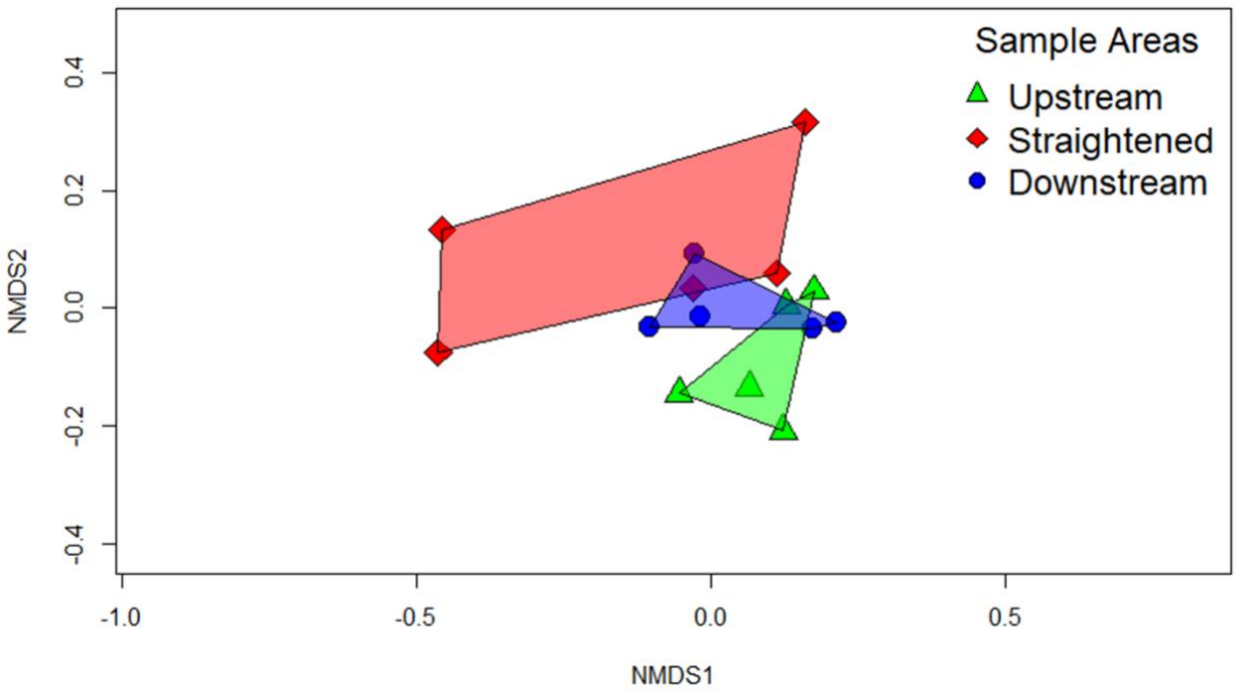


Figure 3. Non-metric multidimensional scaling of the macroinvertebrate abundances sampled across 3 sample areas; upstream, straightened, downstream, from the river Breamish, Northumberland.

A cluster heatmap analysis was conducted to further classify the sites by their macroinvertebrate composition (Figure 4). Results showed the sample sites to be divided into three groups (I, II, III), with group III consisting of Straight 2 & 5, group II consisting of Straight 1, and group I consisting of all the other sites. This classification aligns with the nMDS ordination, with the three furthest straightened points (Figure 3) likely being the sites that make up groups II & III. The macroinvertebrate taxa classified into 7 groups with groups 1-4 & 7 consisting of the single, most abundant taxon (Figure 4). Group 5 consisted of three taxa, *Haliplidae, Limnephilidae,* and *Leptoceridae*, which were abundant in straightened sites. Group 6 is made up of all the other taxa, which varied in abundance across the sites; however, considering the up-and downstream grouped largely to the left of the figure, the abundant species cluster to the left of the figure, containing taxa like *Hydropsychidae*, *Elmidae* (*Limnius*), and *Glossosmatidae,* appear to be characteristic of the upstream and downstream sites.

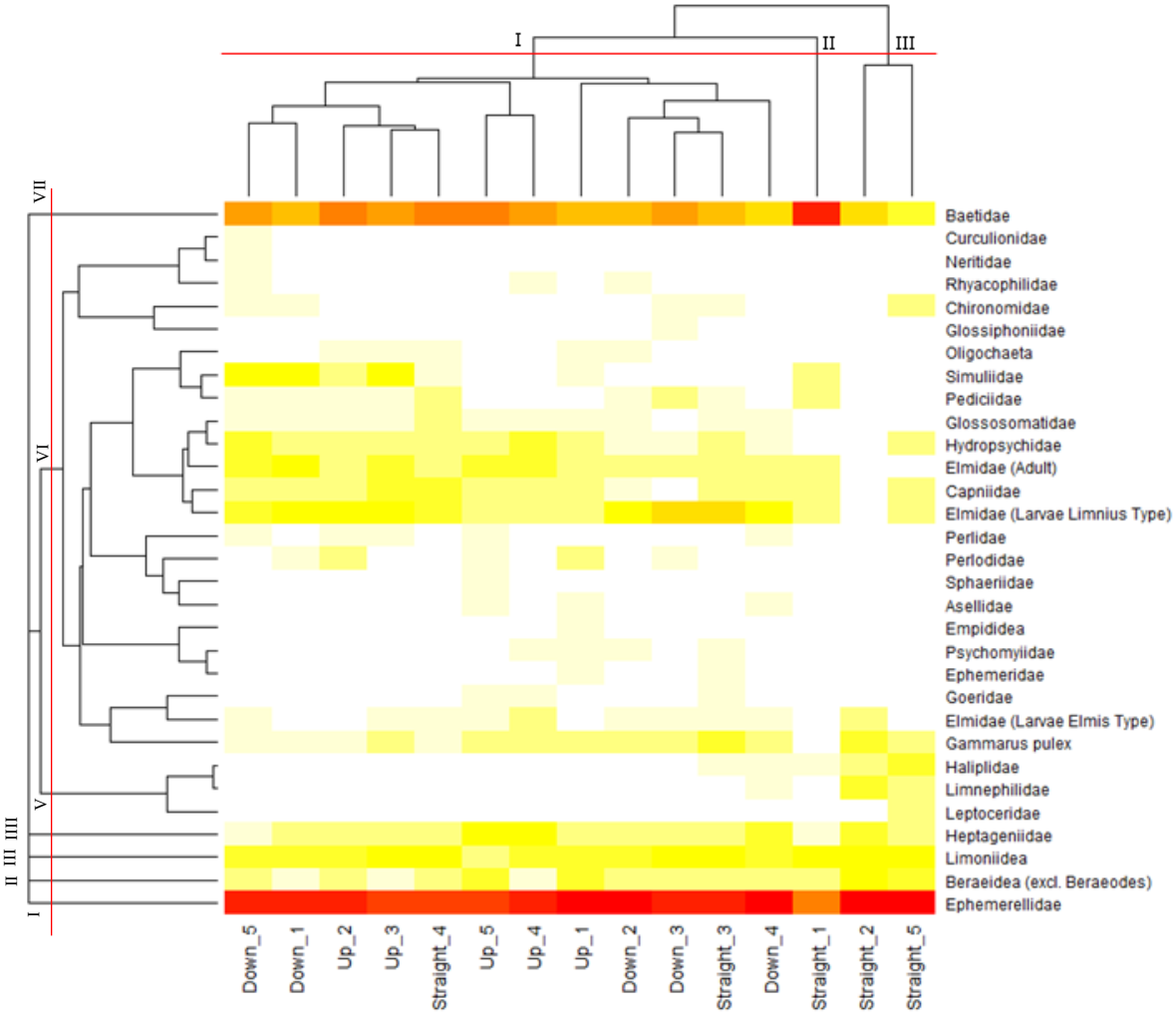


Figure 4. Clustered heatmap diagram of the macroinvertebrate abundances across each sample site. Darker colour indicates higher abundance. Red line indicates clustering cut-off. Groups indicated by roman numerals.

CCA of the macroinvertebrate abundance with the chemical and hydromorphological measurements (minus substrate composition) explained a total of 68.55% of the variation. The first two axes explain 26.51%. However, an ANOVA of the CCA revealed that the environmental variables do not significantly explain the variation in abundance ($F_{10,4}$=0.872, p=0.72). CCA1 is primarily influenced by temperature, ORP, pH and ammonium. While CCA2 was primarily influenced by depth, velocity, pH, and ammonium (see Appendix 2). CCA of the abundances with substrate composition was found to explain a total of 41.53% of the variation, with the first two axes explaining 24.98%. A comparison of the two CCA's eigenvalues indicates the first CCA to have lower eigenvalues across the first few axes but explains a larger total proportion of the variation. As such, the second CCA was seen to have a stronger effect on individual axes but not collectively explain much variation, this suggests some of the individual substrate types likely have a strong influence. CCA1 was mainly influenced by silt, clay, and pebbles. CCA2 was primarily influenced by gravel, cobble, and pebble (see Appendix 3).

Biotic Indices

Little statistically significant differences were found for the taxonomic indices calculated for the macroinvertebrate communities inhabiting the different sample areas. However, on average the straightened sites were observed to be more taxonomically even (1.47) with reduced Shannon diversity (2.24) and Margalef richness (13.20) scores in comparison to the upstream and downstream sites (Table 7). On average the downstream sites exhibited a reduction in diversity scores (2.35), compared to the upstream (2.44). This trend was similarly observed for the calculated BMWP scores, WHPT score, and Ntaxa, with upstream having on average the highest scores, followed by downstream, then straightened (Table 7). EPT scores showed an average decline between sample areas (Table 7), with the lowest score being downstream (78.41) and the highest upstream (82.69). However, it should be noted, that the lowest EPT score of any site was 73% depicting the high abundance of pollution-sensitive taxa. A one-way ANOVA revealed there to be a statistically significant difference ($F_{2,12}$=4.57, p=0.030) in the LIFE scores expressed across the sample areas. A posthoc Tukey HSD test showed a borderline nonsignificant difference (p=0.056, 0.053) between the straightened (8.26) and upstream (8.70) & downstream (8.69) sites. No statistically significant difference was found between upstream and downstream. PSI scores displayed a similar trend (Table 7), with the lowest average score in the straightened (91.39) section, although they did not statistically differ between areas.

Table 7 Average biotic indices using the sampled macroinvertebrates from across the three sample areas. ± standard deviation. Numbers in bold indicate the lowest values between sample areas. Highlighted boxes indicate the highest values between sample areas.

Biotic Indices	Upstream	Straightened	Downstream
Diversity	2.435 ± 0.052	**2.242 ± 0.221**	2.348 ± 0.110
Richness	16 ± 1.140	**13 ± 3.493**	16 ± 1.871
Evenness	**1.336 ± 0.030**	1.474 ± 0.154	1.348 ± 0.051
WHPT Score	69 ± 24.338	**50 ± 9.626**	56 ± 5.469
WHPT Ntaxa	13 ± 1.673	**11 ± 2.608**	12 ± 1.483
WHPT ASPT	5.072 ± 1.093	4.777 ± 0.273	**4.615 ± 0.120**
BMWP Score	85 ± 8.408	**69 ± 14.328**	78 ± 12.329
BMWP Ntaxa	12 ± 1.304	**10 ± 2.074**	11 ± 1.483
BMWP ASPT	3.533 ± 0.350	**2.858 ± 0.597**	3.250 ± 0.514
EPT Score	82.691 ± 5.879	82.084 ± 4.042	**81.671 ± 4.051**
LIFE Score	8.696 ± 0.303	**8.259 ± 0.235**	8.691 ± 0.250
PSI Score	95.701 ± 5.967	**91.388 ± 5.052**	95.904 ± 5.302

Functional Analysis

No statistically significant difference were found between the sample area's functional diversity, richness, or evenness, of macroinvertebrates. Similarly, a PERMANOVA analysis (using 9999 permutations) of the trait-abundance matrix found there to be a borderline nonsignificant difference (pseudo-$F_{2,12}$=1.97, p=0.075) in the abundance of traits expressed at each sample area, with only 24.64% of the variation explained (R^2=0.2464). However, as displayed in Table 8, SIMPER analysis revealed several statistically significant differences ($p<0.05$) in the proportion of different functional traits expressed between sample areas (cumulative results in Appendix 4). For example, the locomotive trait for 'epibenthic burrowing' (LSR5) was expressed significantly more ($p<0.05$) in the straightened area compared to the upstream and downstream (Table 8). The substrate preference for 'mud' (S9) and the 'piercer' feeding habit (FH6) were found to be significantly more exhibited in the straightened sites than the upstream ($p<0.05$), while traits such as 'cemented isolated eggs' (R3), preference for fast '> 50 cm s^{-1}' velocity (CV4), preference for 'flags/boulder/cobble/pebbles' (S1), and 'gravel' (S2) were all found to be expressed significantly more in the upstream sites (Table 8). 'Piercer' (FH6) and 'deposit feeder' (FH2) feeding habit, 'full water swimming' (LSR3), and preference for 'organic detritus/litter' (S8) and 'null' velocity (CV1) were all found to be statistically more expressed in the straightened reach compared to downstream ($p<0.05$). 'Scraper' feeding habit (FH4) was found to be statistically more expressed upstream and downstream compared to the straightened reach ($p<0.05$). Average dissimilarity was highest between the downstream and straightened (0.0011 ± 0.0006),

followed by upstream and straightened (0.001 ± 0.0006), then upstream and downstream (0.0009 ± 0.0006), the degree of standard deviation of these averages likely explains the nonsignificant PERMANOVA result.

Table 8. Statistically significant SIMPER analysis outputs between the macroinvertebrate functional trait abundances between the 3 sampled areas. ava: average abundance in the first group. avb: average abundance in the second group. Highlighted ava & avb indicates the highest abundance displayed by the sample site in the comparison. * $p < 0.05$. ** $p < 0.001$. *** $p < 0.0001$.

Upstream vs Straightened				Straightened vs Downstream				Upstream vs Downstream			
Trait	ava (up)	avb (str)	p	Trait	ava (str)	avb (ds)	p	Trait	ava (up)	avb (ds)	p
FH4	0.94	0.86	*	FH4	0.86	0.95	*	MPS3	1.60	1.49	*
R3	0.91	0.81	***	LSR5	0.25	0.18	*	FH3	0.86	0.75	*
CV4	0.70	0.60	*	CV1	0.34	0.28	*	-	-	-	-
RC1	0.21	0.11	**	D3	0.36	0.34	*	-	-	-	-
S1	0.58	0.50	*	F5	0.47	0.42	*	-	-	-	-
LSR5	0.18	0.25	*	FH2	0.41	0.37	**	-	-	-	-
S2	0.36	0.32	*	FH6	0.05	0.01	*	-	-	-	-
FH6	0.01	0.05	*	S8	0.29	0.25	*	-	-	-	-
F6	0.07	0.09	*	LSR3	0.39	0.35	*	-	-	-	-
S9	0.11	0.13	**	-	-	-	-	-	-	-	-
RS2	0.02	0.01	*	-	-	-	-	-	-	-	-

Fourth Corner Problem

Fourth corner analysis displayed 12 of the 69 traits and 7 of the 18 chemical and hydromorphological variables to have significant associations (Table 9). Notably, velocity was found to be positively correlated with the 'temporarily attached' (LSR7) locomotion trait (r=0.11, p=0.023) and 'filter feeder' (FH5) feeding habit (r=0.11, p=0.029). ORP was found to be negatively correlated to 'living macrophytes' (F4) consumption (r=-0.08, p=0.031) and 'scraper' (FH4) feeding habit (r=-0.09, p=0.029). A preference for slow '<25 cm s^{-1}' (CV2) current velocity was negatively associated with velocity (r=-0.09, p=0.041) and %pebble (r=0.09, p=0.049). The preference for 'silt' (S4) was positively correlated to %clay (r=0.08, p=0.034), but negatively correlated to %pebble (r=-0.09, p=0.032).

Table 9. Fourth corner analysis results display the relationships between macroinvertebrate functional traits and environmental variables. Results highlighted indicate a positive relationship, and blank indicates a negative relationship. * $p < 0.05$. ** $p < 0.001$.

Environment Variable x Trait	Correlation Coefficient	Standard Observed Value	p-value
Width / MPS2	-0.076	-1.954	0.045 *
ORP / MPS2	-0.078	-2.022	0.039 *
Silt / RC1	-0.072	-2.275	0.015 *
Width / AS4	-0.073	-1.846	0.043 *
Clay / AC4	0.079	1.845	0.033 *
Width / LSR1	-0.079	-2.231	0.018 *
ORP / LSR1	-0.080	-2.532	0.015 *
Velocity / LSR7	0.112	2.117	0.023 *
Velocity / F2	0.141	2.353	0.010 **
Width / F4	-0.087	-2.147	0.024 *
ORP / F4	-0.084	-2.111	0.031 *
Nitrate / F9	-0.123	-3.015	0.036 *
ORP / FH4	-0.087	-2.140	0.029 *
Velocity / FH5	0.115	2.411	0.029 *
Clay / S4	0.078	1.857	0.034 *
Pebble / S4	-0.090	-1.798	0.032 *
Velocity / CV2	-0.095	-1.946	0.041 *
Pebble / CV2	-0.091	-1.893	0.049 *

MoRPh Habitat Survey

Due to the omission of invasive non-native macrophytes during the MoRPh surveys, only 13 of the 14 MoRPh indices were calculated (Table 10). ANOVAs found Index 10 (Riparian physical habitat complexity) and Index 12 (Degree of human pressure imposed by bank top land cover) to statistically differ between sample areas. Post-hoc Tukey HSD analysis revealed the statistical differences to occur in Index 10 between upstream and downstream ($p=0.027$) and straightened and downstream ($p=0.027$) and in Index 12 between only the straightened and downstream sites ($p=0.015$). Index 10 was seen to increase on average in the downstream (1.79) sites compared to the upstream and straighten (0.66) sites (Table 10). Index 12 was observed to be highest in the straightened (3) section, then upstream (1.6), followed by downstream (1). While the other indices were found to be nonsignificant, several of them display notable trends across the average index values (Table 10). For example, Index 1 (Number of present/extensive flow types) was on average highest upstream (3) and lowest in the straightened section (2). Index 5, 6, and 7 (Table 10), describe the average riverbed sediment size/class (Index 5 & 6) and extent of bed siltation (Index 7). This displays similar trends with upstream having the biggest average phi units (-2.34, GP) and least siltation (0.63) and the straightened section having the smallest average phi units (0.96, SA) and most siltation (2.2). While downstream fell in between, note the average sediment size/class (-0.48, SA) and extent of siltation (2.06) were closer to the scores produced by the straightened sites. The channel physical habitat complexity (Index 8) was found to be on average lowest in the straightened (2.83) section and highest upstream (3.17), and Index 11 (Riparian vegetation structural complexity) followed a similar trend in averages as the statistically significant Index 10, with a downstream increase (Table 10).

CCA of the macroinvertebrate abundance with MoRPh index values (minus categorical indices) explained a total of 66.15% of the variation. The first two axes explain 43.67%. However, an ANOVA of the CCA revealed that the indices do not significantly explain the variation in abundances ($F_{10,4}=0.782$, $p=0.87$). CCA1 was primarily influenced by indices 1, 8, 12, and 13. CCA2 was primarily influenced by indices 5, 3, 7, and 10 (see Table 2 for codes, see Appendix 5 for graphs).

Table 10. MoRPh River Habitat Survey average indices were calculated for the three sample areas. ± standard deviation. Refer to Table 2 for a description of the individual MoRPh indices. BW: broken waves. RP: rippled. UW: upwelling. GP: gravel/pebble. SA: sand.

MoRPh Indices	Upstream	Straightened	Downstream
Index 1	3 ± 0.71	2 ± 0.71	2.4 ± 0.89
Index 2	BW	RP/UW	RP
Index 3	2 ± 0.71	2.8 ± 1.10	2.6 ± 0.55
Index 4	GP	GP	GP
Index 5	-2.34 ± 1.48	0.96 ± 2.83	-0.48 ± 2.26
Index 6	GP	SA	SA
Index 7	0.63 ± 0.80	2.2 ± 2.74	2.06 ± 1.24
Index 8	3.17 ± 0.70	2.83 ± 0.68	3.08 ± 0.63
Index 9	1.4 ± 0.55	1.4 ± 0.55	1.6 ± 0.89
Index 10	0.66 ± 0.26	0.66 ± 0.23	1.79 ± 0.97
Index 11	4.7 ± 1.25	5.8 ± 1.44	6.1 ± 1.34
Index 12	1.6 ± 1.52	3 ± 0	1 ± 0.61
Index 13	0.06 ± 0.14	0.49 ± 0.65	0 ± 0

RICT Model

Comparisons of the observed and predicted RICT indices, to create EQI values for each index at each site, are reported in Table 11. All BMWP and WHTP indices displayed reduced average EQIs in the straightened reach, with upstream exhibiting the highest average EQIs. BMWP score, NTaxa, ASPT, WHTP score, and NTaxa EQIs suggested that all sites across the sample areas were slightly degraded or worse (Table 11) than expected at a 'pristine' site. Particularly, WHPT score EQIs suggested that two upstream sites (2 & 3), three downstream sites (1, 2, & 3), and four straightened sites (1, 2, 4 & 5) were in 'very degraded' condition. However, WHPT ASPT EQIs were largely in the expected range, with 'slight degradations' upstream at Site 4 & 5, and downstream at Site 4. The straightened reach exhibited the most 'slight degradations' at Sites 3, 4 and 5 (Table 11). LIFE score EQIs were all within the expected range, and many of the site's observed values were increased compared to predicted, but on average LIFE score EQI was reduced in the straightened (0.998) reach and highest downstream (1.071), indicating a greater abundance of pollution-sensitive taxa downstream and upstream of the straightened site (Table 11). This trend was also exhibited by the PSI score EQIs (Table 11), with the average PSI score EQI being lowest in the straightened (1.061) section and highest downstream (1.144). Upstream sites 2 & 3 and downstream sites 1 & 2 displayed 'better than expected'

PSI score EQI values, indicating a much less silted site observed than predicted by RICT (Table 11).

Table 11. Observed/Expected Ratios (Environment Quality Index, EQI) between the observed biotic indices and the predicted/expected biotic indices predicted by RICT. Red indicates Very Degraded (> 0.3), Orange indicates Moderate Degraded (0.49-0.30). Yellow indicates Slightly Degraded (0.69-0.50). White indicates Expected (1.19-0.70). Blue indicates Better than Expected (< 1.2).

Site	BMWP Score	BMWP Ntaxa	BMWP ASPT	WHPT Score	WHPT Ntaxa	WHPT ASPT	LIFE Score	PSI Score
Upstream 1	0.571	0.534	0.582	0.605	0.586	1.036	0.999	1.024
Upstream 2	0.507	0.431	0.592	0.298	0.413	0.725	1.136	1.217
Upstream 3	0.436	0.386	0.520	0.271	0.373	0.732	1.102	1.229
Upstream 4	0.519	0.425	0.510	0.312	0.455	0.686	1.055	1.111
Upstream 5	0.625	0.536	0.634	0.342	0.515	0.665	0.999	0.989
Straight 1	0.347	0.288	0.405	0.204	0.255	0.805	1.034	1.117
Straight 2	0.322	0.279	0.337	0.211	0.284	0.744	0.957	0.990
Straight 3	0.543	0.479	0.568	0.330	0.498	0.665	1.008	1.036
Straight 4	0.412	0.401	0.431	0.273	0.393	0.696	1.027	1.140
Straight 5	0.510	0.451	0.520	0.296	0.438	0.677	0.966	1.021
Downstream 1	0.453	0.398	0.525	0.278	0.384	0.727	1.112	1.213
Downstream 2	0.408	0.372	0.459	0.283	0.396	0.716	1.084	1.202
Downstream 3	0.385	0.366	0.440	0.241	0.324	0.747	1.080	1.153
Downstream 4	0.533	0.482	0.556	0.312	0.465	0.672	0.989	0.995
Downstream 5	0.582	0.521	0.654	0.324	0.461	0.706	1.092	1.159

As seen in Table 12, the RICT calculated Ntaxa EQR values displayed the lowest average in the straightened (0.438) section, with the highest values seen in the upstream (0.531), followed by downstream (0.467). ASPT EQR was suggested to be on average lowest in the downstream (0.724) reach, closely followed by the straightened (0.728) reach, and highest upstream (0.773). The most common Ntaxa condition class calculated by RICTS for the upstream area was 'poor/moderate', and straightened and downstream were both in 'bad' condition (Table 12). The most common ASPT condition class upstream and downstream was 'moderate/poor', with the straightened reach found to be in 'poor/moderate' condition. MINTA condition classes calculated by RICTS found the upstream area to be most commonly in 'poor' condition, while the straightened and downstream sites were in 'bad' condition (Table 12).

Table 12. WHTP Ntaxa and ASPT Environment Quality Ratios classified by RICT for each sample site, as well as the respective Ntaxa, ASPT, and MINTA status classification. Red indicates Bad. Orange indicates Poor. Yellow indicates Moderate. Blue indicates High.

Sample Sites	Ntaxa EQR	ASPT EQR	Ntaxa Class	ASPT Class	MINTA Class
Upstream 1	0.652	1.015	**M**	**H**	**M**
Upstream 2	0.471	0.737	**B**	**M**	**B**
Upstream 3	0.430	0.743	**B**	**M**	**B**
Upstream 4	0.522	0.697	**P**	**P**	**P**
Upstream 5	0.582	0.675	**M**	**P**	**P**
Straightened 1	0.312	0.815	**B**	**M**	**B**
Straightened 2	0.349	0.754	**B**	**M**	**B**
Straightened 3	0.565	0.676	**M**	**P**	**P**
Straightened 4	0.459	0.706	**B**	**P**	**B**
Straightened 5	0.504	0.688	**B**	**P**	**P**
Downstream 1	0.443	0.738	**B**	**M**	**B**
Downstream 2	0.457	0.726	**B**	**M**	**B**
Downstream 3	0.383	0.757	**B**	**M**	**B**
Downstream 4	0.531	0.682	**P**	**P**	**P**
Downstream 5	0.521	0.717	**P**	**P**	**P**
Upstream Average	0.531 ± 0.088	0.773 ± 0.138	**P/M**	**M**	**P**
Straightened Average	0.438 ± 0.105	0.728 ± 0.057	**B**	**P**	**B**
Downstream Average	0.467 ± 0.061	0.724 ± 0.028	**B**	**M**	**B**

Figure 5 displays the results of the Procrustes rotation between the straightened reach's biotic indices predicted by RICT, using the observed environmental input variables and the upstream environmental input variables. This was done to assess the influence of the environmental variables on the predicted indices. Procrustes analysis indicated there to be no significant alignment between the two PCAs of predicted indices ($p>0.05$). Although there was a moderate alignment shown in the correlation coefficient (0.625) and a Procrustes sum of squares (0.605), displaying some level of similarity. Nonsignificant alignment is visualised in Figure 5 with the length of residuals falling mainly around or above the 25% quartile, suggesting a considerable amount of discrepancy between the two PCAs.

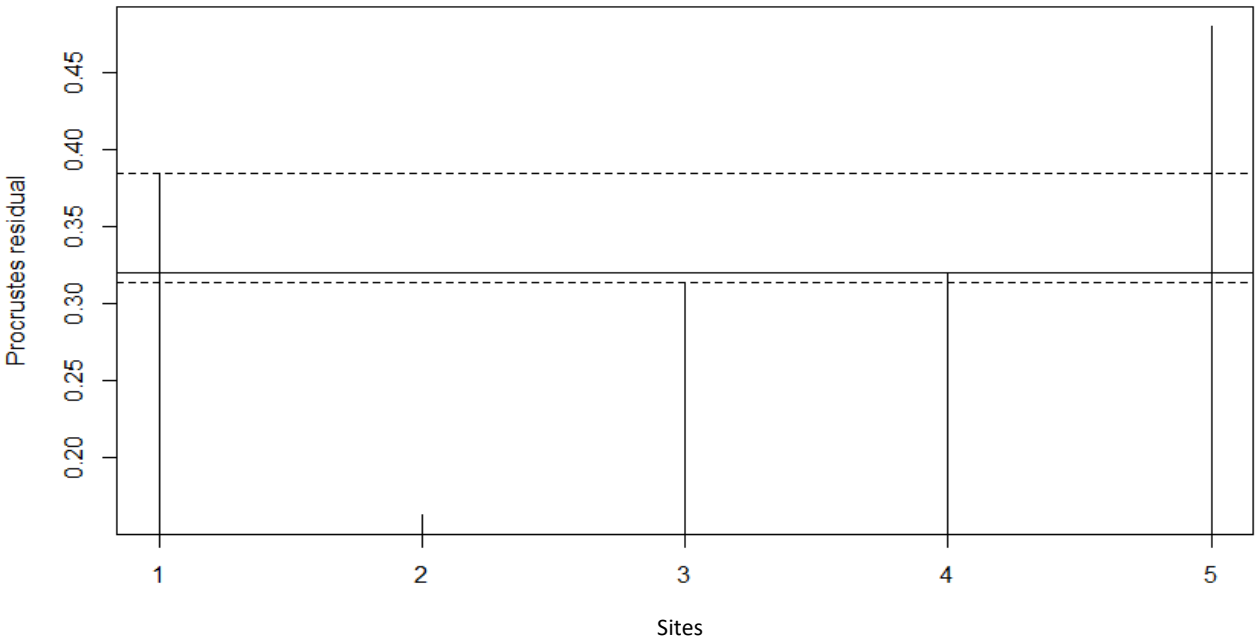


Figure 5. Procrustes errors between the PCAs of predicted straightened sites using observed and proxy environmental variables. Bottom dotted line: 25% quartile, middle blank line: 50% quartile, top dotted line: 75% quartile.

Discussion

The results of this study highlight the high overall ecological health and water quality of the Breamish. It also demonstrates the continued impact that river straightening has on macroinvertebrate communities, river morphology, and habitat features, even ~200 years since the initial modifications occurred. The river straightening drastically alters natural flow regimes and substrate composition, notably an increase in fine sediment, which leads to specific differences in the presence and abundance of certain taxa and functional traits expressed. Further key findings of this research include the observed degradation of water quality from upstream to downstream, as well as the presence of increased impact downstream of the straightening. It is important to note that the present study occurred only over a single season (Spring/Early Summer of 2024) with a relatively low number of samples (n=15). While still providing valuable insight, caution is needed when determining the precise causal relationships between river straightening and its ecological impact.

The present study found the straightened reach to be similar in average channel width but on average deeper, than both the up-and downstream sections (Table 4). This suggests that the straightened reach would likely be characterised by a larger, more uniform cross-sectional area and thus potentially a lower hydraulic roughness (Nittrouer *et al.*, 2012; Vermeulen & Hoitink., 2015) compared to the 'more natural' meandering sections. A larger cross-sectional area facilitates a more consistent water flow, reducing the energy dissipation that would typically occur in a complex channel (Zhou & Endreny., 2020). This results in a slower flow velocity, as observed in this study (Table 4). As expected for a high-gradient wandering channel (Hohensinner *et al.*, 2018), the upstream site's substrate composition was gravel/pebble-dominated. The straightened section showed a marked increase in fine sediments (silt/sand), while the downstream section reverted to a gravel/pebble dominant substrate, with an increase in fine sediments (Table 4). The observed increase in fine sediment in the straightened reach has likely occurred due to the extensive erosion of the flood banks present (Per Obs.), coupled with the slower flow velocity, which reduces the transport capacity of the reach (Hohensinner *et al.*, 2018). The increase in fine sediment present in the downstream reach is likely an artefact of the straightening, which displays the downstream impact of river channelisation (Parker & Andres., 1976; Hohensinner *et al.*, 2018; Heritage & Entwistle., 2020).

Water quality across all sample sites was observed to be high, with nutrient concentrations largely falling within the acceptable limits for aquatic ecosystems (Poikane *et al.*, 2021). No detectable amount of phosphate was present (Table 4). However, average nitrate concentrations were observed to increase from upstream to downstream. Average ammonium was also seen to spike in the straightened reach in comparison to upstream and downstream (Table 4). These findings may be explained by a reduction in hyporheic exchange within the straightened reach. Increased fine sediment limits the movement of water between the channel and hyporheic zone in comparison to coarser substrates (Jin *et al.*, 2019; Shrivastava *et al.*, 2020). Moreover, Mojarrad *et al* (2019) found that increasing flow resulted in increased hyporheic exchange. Therefore, both the observed slower flow and increased fine sediment could likely be principal drivers in the observed findings by reducing the processes of denitrification and ammonium removal. Both conductivity and TDS were seen to increase from upstream to downstream, while ORP decreased (Table 4). One interpretation of these findings could be an increase in dissolved solids, such as minerals and salts, due to the extensive erosion occurring along the flood banks (Adjovu *et al.*, 2022). Moreover, increased organic matter within the system because of erosion and sedimentation (Worral *et al.*, 2015), could likely be reducing the ORP through the consumption of oxygen as the organic matter decomposes.

The results of this research provide supporting evidence that river straightening alters the presence of habitat features and channel morphology (Talbot & Lapointe., 2002; Zhou & Endreny., 2020), even after ~200 years since initial modifications occurred. Through undertaking MoRPh surveys, the straightened section was observed to have the lowest number of present flow types and the lowest value for physical habitat complexity (Table 10). This finding is in accordance with Zhou & Endreny's (2020) research which observed the loss of hydraulic complexity due to river straightening. Decreased hydraulic complexity has been acknowledged to correlate with fewer habitat types and lower biological diversity (Gualtieri *et al.*, 2020). Moreover, hydraulic complexity is known to control streambed topography, and thus influence patterns in hyporheic exchange (Harvey & Bencala., 1993), providing further support to the potential reduction in hyporheic exchange. These results coupled with the observation that the straightened section had the highest degree of human pressure imposed by bank top land cover, as well as the highest degree of channel reinforcements (Table 10), strongly imply

that the anthropogenic modifications are a primary driving force behind alterations to habitat features and channel morphology. Aligning with the visual interpretation of substrate composition, the straightened reach was shown to be characterised by the lowest alluvial bed material size in phi units and the highest extent of bed siltation. Downstream sites were observed to have a lower bed material size and higher extent of siltation than upstream sites (Table 10), strongly suggesting the presence of a downstream impact caused by river straightening. These findings further support the assumed impact that river straightening has on river flow, and thus substrate composition, through reduced sediment transport out of the system (Hohensinner *et al.*, 2018; Heritage & Entwistle., 2020).

Under low flow conditions, the total number of macroinvertebrates (Figure 2), Margalef richness and Shannon diversity in the straightened section were observed to be lower, and Pielou's evenness to be higher than those found upstream and downstream (Table 7), consistent with the findings of previous studies (Negishi *et al.*, 2002; Horsák *et al.*, 2009; Käiro *et al.*, 2017). While these findings display the negative impact of river straightening, the high overall ecological health and water quality of the Breamish is especially evident in the high EPT and LIFE scores observed across all three sample areas (Table 7). EPT scores ranged from 73 - 89% across all sample sites displaying the high percentage of environmentally sensitive taxa present (Kitchin., 2005). LIFE scores ranged from 8.00 - 9.08 across all sample sites, which similarly depicts typical results of high-gradient headwaters (Extence *et al.*, 1999). Despite high overall EPT and LIFE values, differences were observed between specific sample areas which reflect the impact of the river straightening (Table 7). The EPT scores declined from an average of 82.7% to 81.5% to 78.4% from upstream to downstream, which suggests a decline in water quality in relation to the percentage of pollution-sensitive taxa. Average LIFE scores were lowest in the straightened section compared to upstream and downstream (Table 7), reflecting the observed reduced flow velocity. Despite the visual interpretation of substrate composition suggesting a marked increase in fine sediment, the PSI index suggests all sample areas to be largely silt-free (Extence *et al.*, 2013) with PSI scores ranging from 86.7-100, across all sample sites. However, as observed with the LIFE scores, average PSI scores were seen to be lowest in the straightened reach (Table 7), reflecting the visually observed increase in fine sediment.

The ecological effect of river straightening has often been cited to cause an increase in flow velocity within the impacted section (Brooker., 1985; Kesel., 2003; Knighton., 2014; Hohensinner *et al*., 2018). This is suggested to induce a shift in community composition towards that more commonly found in upstream reaches, termed the "rhithralization effect" (Jungwirth *et al*., 2000). However, the findings of the current study do not accord to the presence of such an effect for the studied straightened section of the Breamish. Macroinvertebrate compositions displayed specific differences in the abundance of certain taxa, which reflected a preference for an overall slower flow velocity in the modified reach (Figures 3 & 4; Table 6). Taxa such as crawling water beetles, *Haliplidae*, and Limoniid crane flies, *Limoniidae*, were found to be statistically more abundant in the straightened section compared to upstream and downstream. *Haliplidae* is acknowledged to inhabit slow-flowing lentic habitats (Roughley., 2000) and similarly, *Limoniidae*, which was previously a subfamily of *Tipulidae*, have been observed to occur in abundance in slow-flowing to standing water (Stubbs., 2021). Although not found to be statistically significant other lentic taxa such as *Leptoceridae* (Waringer & Graf., 2011), *Limnephilidae* (Hiley., 1978), and *Pediciidae* (Stubbs., 2021) were found to be more abundant in the modified section (see Appendix 1). Existing research has proposed certain channelization measures to exhibit a "potamalization effect" (Jungwirth *et al*., 2003), where there is a shift in composition to that which would occur in lower reaches under slower flow velocity; although, this is suggested to largely occur upstream of impoundments, such as dams and weirs (Schmutz & Moog., 2018), and not river straightening. The results of this study therefore indicate the presence of a "potamalization effect" within the straightened reach. However, it is highly evident from the extent of channel incision and flood bank erosion (Per Obs.) that under peak flow conditions, and likely during the decades following the initial channelisation, the straightened reach was/is exposed to increased, yet highly variable flow velocities.

The upstream and downstream sample areas were found to have significantly more abundant rheophilic taxa, such as *Hydropsychidae* (Boon., 1988), *Glossosmatidae* (Nijboaer., 2004), *Perlodidae* (Boon., 1988), *and Elmidae* (White., 2009), compared to the straightened site (Table 6). This result concurs with the findings of Boon (1988) who found reductions in *Perlodidae* and net-spinning Trichoptera (e.g. *Hydropsychidae)* abundance with increasing channel regulation. Nijboaer (2003) outlines the importance that *Glossosmatidae* has as a characteristic species of unimpacted streams. The increased

abundance of *Elmidae*, the riffle beetle, clearly displays the presence of fast-flowing shallow areas (White., 2009). As seen in Figure 3, there was a wider variation of taxa composition for straightened sites compared to up and downstream. At first, this appears counterintuitive, but this likely reflects sections of 're-naturalisation' occurring along the straightened reach (Figure 6), possibly in response to erosion. Lane (1947) displayed the occurrence of long-term changes in straightened erodible channels, but non-erodible channels could not adjust.



Figure 6. Drone imagery of the straightened reach taken by Matt Carr (30/08/2024). Yellow circle highlights clear 're-naturalisation' occurring erosion/deposition associated meander, riffle/pool. Dark blue circle highlights areas where river is breaching onto the floodplain, very minimal. Additions to image by author.

The RICT model indicates all sample sites to have slightly degraded or worse EQIs for all indices except WHPT ASPT, LIFE scores, and PSI scores (Table 11). Likewise, EQRs computed by the RICT classification, particularly the MINTA scores, suggested all sites apart from Upstream 1 (moderate condition) to be in 'poor' or 'bad' condition (Table 12). These are interesting results considering the upstream and downstream sites were expected to produce EQIs and EQRs that show a more 'natural' environment. This could likely be explained by the presence of flood embankments at the upstream and downstream sample areas, albeit with no river straightening, which could be negatively influencing the macroinvertebrate composition. Existing literature has seldom examined the direct effects of flood banks on fluvial taxa within the river channel; however, it could be postulated that the floodplain disconnection could be reducing the observed

composition (Rideout *et al.*, 2022) in comparison to 'pristine' sites predicted by RICT, that would have fully functional lateral connectivity.

For riverine systems, river flow is often viewed as the 'master variable' (Power *et al.*, 1995). The interaction between river flow and the channel morphology results in variable patterns in river hydraulic conditions (Zhou & Endreny., 2020). River flow is also widely acknowledged to be a driver of substrate composition, with slower flows associated with finer sediments and faster flows associated with coarser sediments (McCabe., 2011). As such the biological communities inhabiting fluvial systems are adapted to flow regimes and channel characteristics via functional attributes, such as velocity and substrate preference, feeding, locomotion, and reproduction habits. While the functional analysis in this study found no significant difference between functional diversity, richness, or evenness, statistically significant differences in certain functional attributes expressed within each sample area were observed (Table 8). In comparison to upstream and downstream, the functional trait 'epibenthic burrowing' (LSR5) was found to be significantly more expressed in the straightened reach. This finding accords with Azovsky *et al* (2022) who found epibenthic burrowing taxa to prevail in fine silty substrates, while interstitial burrowing taxa prevailed in coarser sediments. Moreover, the substrate preference for 'mud' (S9) was more abundant, compared to upstream, and 'deposit feeding' (FH2) and the substrate preference for 'detritus' (S8) were also more abundant in the straightened reach, compared to downstream. The upregulation of these traits highlights the presence of increased fine sediment within the straightened section. This is further supported by the increase in traits related to the presence of coarser sediment (S1 & S2) in the upstream sites, as well as the increase in 'scraper' feeding habitat (FH4) both upstream and downstream. The 'scraper' feeding habitat refers to organisms which feed on algae (Szałkiewicz *et al.*, 2022), thus requiring the presence of rocks and coarser sediments on which algae grow.

Several differences in attributes related to river flow were also observed (Table 8). Preference for 'null' flow velocity (CV1) and the locomotion trait 'full water swimming' (LSR3) was found to be significantly more abundant in the straightened section compared to downstream, while preference for 'fast' velocity (CV4) and the reproductive trait 'cemented isolated eggs' (R3) were more abundant upstream than within the straightened section. The ability to sustain full water swimming in a dynamic high-flow

environment greatly increases the energy requirement on biota, as such full water swimming is often expressed by taxa inhabiting slow-flowing environments where water movement is gentle enough to support this locomotion (Statzner *et al.*, 2008). Conversely, the production of 'cemented isolated eggs' is an adaptation to a fast-flowing environment (Forio *et al.*, 2023). In such habitats, strong flow can easily wash away loose and unprotected eggs, therefore by cementing eggs, taxa ensure the likelihood of successful development via anchoring eggs to the substrate (Statzner *et al.*, 2008). It could also be postulated that this trait also further indicates the presence of increased coarser sediments, as hard substrata are required for the eggs to be cemented.

This study found several interesting fourth-corner relationships (Table 9). The locomotive trait 'temporally attached', and the feeding habitat 'filter feeder' were found to be positively correlated to flow velocity. Forio *et al* (2023) found similarly that the presence of organisms that were temporally attached increased with increasing flow velocity, as they resisted drifting. Filter-feeding organisms have evolved to typically occur in sites of intense particle transport (Szarek-Gwiazda *et al.*, 2022), such as high-velocity areas of the streams. Furthermore, a preference for 'slow' velocity was found to be negatively associated with velocity and the %pebble. This displays that limnophilic organisms are associated with slower flows, and thus typically finer sediments. A preference for 'silt' was positively associated with %clay, but negatively associated with %pebble, directly displaying the effects of substrate composition on taxa functional attributes.

There are some potential caveats when interpreting the present study's findings. Notably, a drawback of using RICT predictions for a physically modified environment is that RICT utilises the modified environmental data to predict the macroinvertebrate assemblages expected at each site under 'pristine' conditions (Per Coms, Russell Barber - EA, 2024). As such, modifications that are affecting environmental conditions will influence the predicted macroinvertebrate community. For instance, the straightened reach was characterized by a much finer sediment, which could therefore slant the macroinvertebrate predictions towards taxa that have a preference or tolerance to fine sediment. This may lead to an overestimation of the ecological quality of the site (Per Coms). This was investigated through a Procrustes rotation of the original straightened site's predicted indices and an additional set of predicted indices, using the upstream

environmental data. The results of the rotation highlighted a non-significant alignment between the two sets of predicted indices (Figure 5), confirming that the use of the modified environmental data does influence the predicted assemblage. This finding underscores the importance of considering the physical habitat alterations when utilising RICT and highlights the need for caution when interpreting results. Yet, it could be argued that considering only 1% of the UK's river system is unmodified (Jones *et al.*, 2019), it is somewhat implausible to distinguish modified from unmodified environmental conditions, further necessitating the need for a more nuanced approach to interpreting RICT model outputs. A second caveat in this study is the reliance on only a single season of data, as well as the low number of samples. Studies over only a single season greatly limit the temporal scope of the research (Zhao *et al.*, 2024) and fail to capture the full range of environmental and biological variations that occur throughout the year, leading to largely incomplete results. A low number of samples decreases the statistical power of the research and often leads to a higher likelihood of Type II errors (Serdar *et al.*, 2021), which was likely observed in this study. As well as this, too few samples also magnify the impact of outliers, therefore potentially skewing the results and reducing the reliability of the study (Konietschke *et al.*, 2020).

A novel aspect of the present study is that it is the first research of its kind to show the effects of river straightening, following ~200 years since the modification occurred. This highlights the highly sustained impact of river straightening on fluvial ecology, particularly in reaches that are no longer maintained and left to function like natural systems. This research also benefits from an integrated approach to link the effects of river straightening to physical channel characteristics, habitat features, macroinvertebrate composition, and functional traits. This type of comprehensive approach creates a more holistic understanding of the ecological impact of river straightening. Furthermore, this study also provides a good baseline for further research following the planned restoration of the straightened reach. River restoration initiatives have often been cited to seldom conduct the appropriate pre-restoration monitoring (England., 2021), making it difficult to judge the actual ecological success of restoration schemes. As such, the results of this research can be compared following restoration to deduce changes in ecological and environmental quality.

For the expected long-term changes, particularly following the planned river restoration scheme (Figure 7), it can be assumed that the observed 'potamalization effect' in the straightened section will worsen, due to the diversion of river flow into the constructed meandered channel and construction of leaky dams (Figure 7). These alterations will cause the flow to reduce in the straightened reach and sedimentation to likely increase. Following the recolonisation of macroinvertebrates, the assemblage of the constructed meander channel will likely consist of a rheophilic community, and thus the associated functional trait, as observed up- and downstream, if not better when considering the re-establishment of lateral connectivity to the floodplain (Gallardo *et al.*, 2008; Rideout *et al.*, 2022). The increase in floodplain connectivity will also likely reduce the observed downstream impact of sedimentation by allowing fine sediments to naturally settle on the floodplain (Asselamn & Middelkoop., 1995), potentially increasing macroinvertebrate richness and diversity (Rideout *et al.*, 2022). The restoration will likely cause an overall biological net gain. Furthermore, the restoration will also reduce downstream flood impact through re-connection with the floodplain, allowing for more water to be stored during peak flow, benefiting downstream human communities (Dixon *et al.*, 2016). This is important considering the increase in extreme weather events due the anthropogenic climate change (Mirza., 2003), as highlighted in this study being hindered by abnormally high rainfall and river levels, during the spring/early summer of 2024 (Chan *et al.*, 2024; Turner *et al.*, 2024).



Figure 7. Drone imagery of the existing straightened reach of the Breamish (top) and proposed landscape (bottom) following the planned river restoration. Note (bottom) – additional of leaky dams in straightened reach, and creation of wetland on left floodplain. Images created by Neil Williams (AECOM).

In terms of future research, it would be useful to extend the current findings by examining the effects of river straightening across a greater longitudinal scale to deduce the effect of seasonal variations. This research should also be replicated at the same site following the restoration, ideally with annual surveys, to assess the change in ecological and water quality and add to the growing body of literature related to river restoration's effect on fluvial and riparian ecology.

Conclusion

In conclusion, this study provides evidence of the long-term ecological impact of river straightening on the Breamish. It demonstrates alterations in channel morphology, habitat complexity, and macroinvertebrate taxonomic community and functional structures, even after approximately two centuries. The results show shifts towards slow velocities and increased fine sediment deposition within the straightened reach, which aligns with distinct shifts in taxa and functional attributes. While the ecological health and water quality are high, the study highlights the impact of historical river straightening and its continued effect on fluvial ecology. Following the restoration, regular long-term monitoring is needed on an annual basis, to fully understand ecological outcomes, and guide effective management of future river restoration initiatives.

References

Ab Hamid, S. and Rawi, C.S.M., 2017. Application of aquatic insects (Ephemeroptera, Plecoptera and Trichoptera) in water quality assessment of Malaysian headwater. Tropical life sciences research, 28(2), p.143.

Adams, M., 1999. Beyond the Pale: some thoughts on the later prehistory of the Breamish valley. Bevan (ed), pp.111-22.

Adjovu, G.E., Stephen, H. and Ahmad, S., 2022. Monitoring of Total Dissolved Solids Using Remote Sensing Band Reflectance and Salinity Indices: A Case Study of the Imperial County Section, AZ-CA, of the Colorado River. In World Environmental and Water Resources Congress 2022 (pp. 1132-1145).

Arnaud, F., Schmitt, L., Johnstone, K., Rollet, A.J. and Piégay, H., 2019. Engineering impacts on the Upper Rhine channel and floodplain over two centuries. Geomorphology, 330, pp.13-27.

Asselman, N.E. and Middelkoop, H., 1995. Floodplain sedimentation: quantities, patterns and processes. Earth surface processes and landforms, 20(6), pp.481-499.

Azovsky, A.I., Chertoprud, E.S. and Garlitska, L.A., 2022. Community-level spatiotemporal synchrony: new metric and application to White Sea meiobenthic harpacticoids. Marine Ecology Progress Series, 698, pp.55-68.

Bendary, R.E., Goher, M.E. and El-Shamy, A.S., 2023. Taxonomic and functional diversity of macroinvertebrates in sediment and macrophyte habitats: A case study, the Ibrahimia Canal, Nile River, Egypt. The Egyptian Journal of Aquatic Research, 49(2), pp.129-135.

Boon, P.J., 1988. The impact of river regulation on invertebrate communities in the UK. Regulated Rivers: Research & Management, 2(3), pp.389-409.

Bridge, J.S., 2003. Rivers and floodplains: forms, processes, and sedimentary record. John Wiley & Sons.

Brooker, M.P., 1985. The ecological effects of channelization. The Geographical Journal, 151(1), pp.63-69.

Brookes, A. 1988. Channelized rivers: perspectives for environmental management. Wiley-Interscience. ISBN 0-471-91979-9.

Brookes, A., Gregory, K.J. and Dawson, F.H., 1983. An assessment of river channelization in England and Wales. Science of the total environment, 27(2-3), pp.97-111.

Buss, D.F., Carlisle, D.M., Chon, T.S., Culp, J., Harding, J.S., Keizer-Vlek, H.E., Robinson, W.A., Strachan, S., Thirion, C., and Hughes, R.M., 2015. Stream biomonitoring using macroinvertebrates around the globe: a comparison of large-scale programs. Environmental monitoring and assessment, 187, pp.1-21.

Chan, W., Magee, E., Smith, R., Crane, E. and Clemas, S., 2024. Hydrological summary for the United Kingdom: April 2024.

Chessell, A., 2014. Breamish and Till: From Source to Tweed. Lulu.com.

Chesters, R.K., 1980. Biological monitoring working party: the 1978 National Testing Exercise (Technical memorandum). Water Data Unit, Department of the Environment, London.

Chevene, F., Doleadec, S. and Chessel, D., 1994. A fuzzy coding approach for the analysis of long-term ecological data. Freshwater Biology, 31(3), pp.295-309.

Dietrich, W.E. and Smith, J.D., 1983. Influence of the point bar on flow through curved channels. Water Resources Research, 19(5), pp.1173-1192.

Dixon, S.J., Sear, D.A., Odoni, N.A., Sykes, T. and Lane, S.N., 2016. The effects of river restoration on catchment scale flood risk and flood hydrology. Earth Surface Processes and Landforms, 41(7), pp.997-1008.

Dolédec, S., Castella, E., Forcellini, M., Olivier, J.M., Paillex, A. and Sagnes, P., 2015. The generality of changes in the trait composition of fish and invertebrate communities after flow restoration in a large river (French Rhône). Freshwater Biology, 60(6), pp.1147-1161.

Dray, S. and Legendre, P., 2008. Testing the species traits–environment relationships: the fourth-corner problem revisited. Ecology, 89(12), pp.3400-3412.

Dray, S. and Siberchicot, M.A., 2017. Package 'ade4'. Université de Lyon: Lyon, France.

Elosegi, A. and Sabater, S., 2013. Effects of hydromorphological impacts on river ecosystem functioning: a review and suggestions for assessing ecological impacts. Hydrobiologia, 712, pp.129-143.

England, J., Angelopoulos, N., Cooksley, S., Dodd, J., Gill, A., Gilvear, D., Johnson, M., Naura, M., O'Hare, M., Tree, A. and Wheeldon, J., 2021. Best practices for monitoring and assessing the ecological response to river restoration. Water, 13(23), p.3352.

Environment Agency. 2019. Walley Hawkes Paisley Trigg (WHPT) index of river invertebrate quality and its use in assessing ecological status. Environment Agency.

Environment Agency. 2022 reprint 2003. River Habitat Survey Guidance Manual, GOV.UK. Available at: https://www.gov.uk/government/publications/river-habitat-survey-guidance-manual

Environment Agency. 2022a. Ecological Services Framework 3 (EcoSF3): AECOM Harehope River Restoration Project Form. Environment Agency. Available at: https://bidstats.uk/tenders/2022/W14/772342062

EU LIFE., 2021. Life 3.0 - life project public page. Available at: https://webgate.ec.europa.eu/life/publicWebsite/project/LIFE20-NAT-UK-000277/water-and-disturbance-environmental-restoration-on-the-northumbrian-coast

Extence, C.A., Balbi, D.M. and Chadd, R.P., 1999. River flow indexing using British benthic macroinvertebrates: a framework for setting hydroecological objectives. Regulated Rivers: Research & Management: An International Journal Devoted to River Research and Management, 15(6), pp.545-574.

Extence, C.A., Chadd, R.P., England, J., Dunbar, M.J., Wood, P.J. and Taylor, E.D., 2013. The assessment of fine sediment accumulation in rivers using macro-invertebrate community response. River Research and Applications, 29(1), pp.17-55.

Feio, M.J., Hughes, R.M., Callisto, M., Nichols, S.J., Odume, O.N., Quintella, B.R., Kuemmerlen, M., Aguiar, F.C., Almeida, S.F., Alonso-EguíaLis, P. and Arimoro, F.O., 2021. The biological assessment and rehabilitation of the world's rivers: An overview. Water, 13(3), p.371.

Fenn, M., 2018. River Ecology and Urban Pollution. Urban Pollution: Science and Management, pp.173-186.

Forio, M.A.E., Goethals, P.L., Lock, K., Nguyen, T.H.T., Damanik-Ambarita, M.N., Dominguez-Granda, L. and Thas, O., 2023. Determining Tipping Points and Responses of Macroinvertebrate Traits to Abiotic Factors in Support of River Management. Biology, 12(4), p.593.

Frainer, A., Polvi, L.E., Jansson, R. and McKie, B.G., 2018. Enhanced ecosystem functioning following stream restoration: The roles of habitat heterogeneity and invertebrate species traits. Journal of Applied Ecology, 55(1), pp.377-385.

Gallardo, B., García, M., Cabezas, Á., González, E., González, M., Ciancarelli, C. and Comín, F.A., 2008. Macroinvertebrate patterns along environmental gradients and hydrological connectivity within a regulated river-floodplain. Aquatic Sciences, 70, pp.248-258.

Gamble, J. 2023. River Invertebrate Classification Tool (RICT). Freshwater Biological Association. Available at: https://www.fba.org.uk/rivpacs-and-rict/river-invertebrate-classification-tool

Gregory K 2001. Conservation: waterways, in Smelser, N., Baltes, P. (Eds.), International Encyclopedia of the Social & Behaviour Sciences. Pergamon Press, Oxford, UK.

Gualtieri, C., Abdi, R., Ianniruberto, M., Filizola, N. and Endreny, T.A., 2020. A 3D analysis of spatial habitat metrics about the confluence of Negro and Solimões rivers, Brazil. Ecohydrology, 13(1), p.e2166.

Gurnell, A.M. and Downs, P.W., 2021. The legacy of river channel modification in wadeable, lowland rivers: exploring overdeep rivers in England. Earth Surface Processes and Landforms, 46(15), pp.3016-3025.

Harvey, J.W. and Bencala, K.E., 1993. The effect of streambed topography on surface-subsurface water exchange in mountain catchments. Water resources research, 29(1), pp.89-98.

Heritage, G. and Entwistle, N., 2020. Impacts of river engineering on river channel behaviour: Implications for managing downstream flood risk. Water, 12(5), p.1355.

Hiley, P.D., 1978. Some aspects of the life histories of Limnephilidae (Trichoptera) related to the distribution of their larvae. In Proceedings of the 2nd International Symposium on Trichoptera: University of Reading, England, 25–29 July 1977 (pp. 297-301). Dordrecht: Springer Netherlands.

Hohensinner, S., Habersack, H., Jungwirth, M. and Zauner, G., 2004. Reconstruction of the characteristics of a natural alluvial river–floodplain system and hydromorphological changes following human modifications: the Danube River (1812–1991). River Research and Applications, 20(1), pp.25-41.

Hohensinner, S., Hauer, C. and Muhar, S., 2018. River morphology, channelization, and habitat restoration. Riverine ecosystem management: science for governing towards a sustainable future, pp.41-65.

Isom, B.G., 1978. Benthic macroinvertebrates. Methods for the assessment and prediction of mineral mining impacts on aquatic communities: A review and analysis. US Fish and Wildlife Service, Harpers Ferry, West Virginia, pp.67-68.

Jin, G., Chen, Y., Tang, H., Zhang, P., Li, L. and Barry, D.A., 2019. Interplay of hyporheic exchange and fine particle deposition in a riverbed. Advances in Water Resources, 128, pp.145-157.

Jones, J., Börger, L., Tummers, J., Jones, P., Lucas, M., Kerr, J., Kemp, P., Bizzi, S., Consuegra, S., Marcello, L. and Vowles, A., 2019. A comprehensive assessment of stream fragmentation in Great Britain. Science of the total environment, 673, pp.756-762.

Jungwirth M, Haidvogl G, Moog O, Muhar S, Schmutz S (2003) Angewandte Fischökologie an Fließgewässern. Facultas Universitätsverlag, Wien, p 547

Jungwirth M, Muhar S, Schmutz S (2000) Fundamentals of fish ecological integrity and their relation to the extended serial discontinuity concept. Hydrobiologia 422/423:85–97

Kail, J., Brabec, K., Poppe, M. and Januschke, K., 2015. The effect of river restoration on fish, macroinvertebrates and aquatic macrophytes: A meta-analysis. Ecological Indicators, 58, pp.311-321.

Kennedy, T.L. and Turner, T.F., 2011. River channelization reduces nutrient flow and macroinvertebrate diversity at the aquatic-terrestrial transition zone. Ecosphere, 2(3), pp.1-13.

Kenney, M.A., Sutton-Grier, A.E., Smith, R.F. and Gresens, S.E., 2009. Benthic macroinvertebrates as indicators of water quality: The intersection of science and policy. Terrestrial Arthropod Reviews, 2(2), p.99

Kesel, R.H., 2003. Human modifications to the sediment regime of the Lower Mississippi River flood plain. Geomorphology, 56(3-4), pp.325-334.

Kitchin, P.L., 2005. Measuring the amount of statistical information in the EPT index. Environmetrics: The official journal of the International Environmetrics Society, 16(1), pp.51-59.

Knighton, D., 2014. Fluvial forms and processes: a new perspective. Routledge.

Konietschke, F., Schwab, K. and Pauly, M., 2021. Small sample sizes: A big data problem in high-dimensional data analysis. Statistical Methods in Medical Research, 30(3), pp.687-701.

Kozarek, J.L., Hession, W.C., Dolloff, C.A. and Diplas, P., 2010. Hydraulic complexity metrics for evaluating in-stream brook trout habitat. Journal of Hydraulic Engineering, 136(12), pp.1067-1076.

Lamouroux, N., Dolédec, S. and Gayraud, S., 2004. Biological traits of stream macroinvertebrate communities: effects of microhabitat, reach, and basin filters. Journal of the North American Benthological Society, 23(3), pp.449-466.

Land Use., 2010. Northumberland Landscape Character Assessment. Land Use Consultants.

Lane, E.W., 1947. The effects of cutting bends in rivers. Proceedings of the Third Hydraulic Conference, Bulletin No. 31. University of Iowa, IA.

Lau, J.K., Lauer, T.E. and Weinman, M.L., 2006. Impacts of channelization on stream habitats and associated fish assemblages in east central Indiana. The American Midland Naturalist, 156(2), pp.319-330.

Legendre, P., Galzin, R. and Harmelin-Vivien, M.L., 1997. Relating behaviour to habitat: solutions to the fourth-corner problem. Ecology, 78(2), pp.547-562.

McCabe, D. J. (2011) Rivers and Streams: Life in Flowing Water. *Nature Education Knowledge* 3(10):19

Meek, J., Berrell, J., Mulford, T., Shiozawa, D.K. and Nelson, C.R., 2023. Aquatic insect communities remain impaired following large-scale river restoration. Western North American Naturalist, 83(2), pp.207-219.

Menezes, S., Baird, D.J. and Soares, A.M., 2010. Beyond taxonomy: A review of macroinvertebrate trait-based community descriptors as tools for freshwater biomonitoring. Journal of Applied Ecology, 47(4), pp.711-719.

Mirza, M.M.Q., 2003. Climate change and extreme weather events: can developing countries adapt?. Climate policy, 3(3), pp.233-248.

Modular River Survey. 2019. The MoRPh Survey: A modular river physical survey for citizen scientists, Field Guide. Modular River Survey. Available at: http://modularriversurvey.org/wp-content/uploads/morph-field-guide.pdf

Mojarrad, B.B., Betterle, A., Singh, T., Olid, C. and Wörman, A., 2019. The effect of stream discharge on hyporheic exchange. Water, 11(7), p.1436.

Murray-Bligh, J. 2023. RICT User Guide (v111). Freshwater Biological Association.

Murray-Bligh, J., 1999. Procedures for collecting and analysing macro-invertebrate samples. Environment Agency.

Murray-Bligh, J., Griffiths, M. and Forshaw, M., 2022. Freshwater biology and ecology handbook. Foundation for Water Research & Freshwater Biological Association, pp.153-157.

Nakano, D. and Nakamura, F., 2008. The significance of meandering channel morphology on the diversity and abundance of macroinvertebrates in a lowland river in Japan. Aquatic Conservation: Marine and Freshwater Ecosystems, 18(5), pp.780-798.

Nijboer, R., 2004. The ecological requirements of Agapetus fuscipes Curtis (Glossosomatidae), a characteristic species in unimpacted streams. Limnologica, 34(3), pp.213-223.

Nittrouer, J.A., Shaw, J., Lamb, M.P. and Mohrig, D., 2012. Spatial and temporal trends for water-flow velocity and bed-material sediment transport in the lower Mississippi River. Bulletin, 124(3-4), pp.400-414.

Oksanen, J., Blanchet, F.G., Kindt, R., Legendre, P., Minchin, P.R., O'Hara, R.B., Simpson, G.L., Solymos, P., Stevens, M.H.H., Wagner, H. and Oksanen, M.J., 2013. Package 'vegan'. Community ecology package, version, 2(9), pp.1-295.

Palmer, M.A., Bernhardt, E.S., Allan, J.D., Lake, P.S., Alexander, G., Brooks, S., Carr, J., Clayton, S., Dahm, C.N., Follstad Shah, J. and Galat, D.L., 2005. Standards for ecologically successful river restoration. Journal of applied ecology, 42(2), pp.208-217.

Parker, G. and Andres, D., 1976. Detrimental effects of river channelization. In IN: RIVERS'76, PROC. SYMP. ON INLAND WATERWAYS FOR NAVIGATION, FLOOD CONTROL AND WATER DIVERSIONS; 3RD ANNUAL SYMP. OF THE WA (Vol. 2, pp. 1248-1266).

Pawley, S., Dobson, M. and Fletcher, M., 2011. Guide to British freshwater macroinvertebrates for biotic assessment (pp. 80-pp).

Poikane, S., Várbíró, G., Kelly, M.G., Birk, S. and Phillips, G., 2021. Estimating river nutrient concentrations consistent with good ecological condition: More stringent nutrient thresholds needed. Ecological Indicators, 121, p.107017.

Power, M.E., Sun, A., Parker, G., Dietrich, W.E. and Wootton, J.T., 1995. Hydraulic food-chain models. BioScience, 45(3), pp.159-167.

Resh, V.H., Norris, R.H. and Barbour, M.T., 1995. Design and implementation of rapid assessment approaches for water resource monitoring using benthic macroinvertebrates. Australian Journal of Ecology, 20(1), pp.108-121.

Rideout, N.K., Compson, Z.G., Monk, W.A., Bruce, M.R., Hajibabaei, M., Porter, T.M., Wright, M.T. and Baird, D.J., 2022. Environmental filtering of macroinvertebrate traits influences ecosystem functioning in a large river floodplain. Functional Ecology, 36(11), pp.2791-2805.
Roughley, R.E., 2000. Haliplidae Aubé, 1836. American beetles, 1, pp.138-143.

Schmera, D., Podani, J., Heino, J., Erős, T. and Poff, N.L., 2015. A proposed unified terminology of species traits in stream ecology. Freshwater Science, 34(3), pp.823-830.

Schmutz, S. and Moog, O., 2018. Dams: ecological impacts and management. Riverine ecosystem management: Science for governing towards a sustainable future, pp.111-127.

Sear, D., Wilcock, D., Robinson, M. and Fisher, K., 2012. River channel modification in the UK. In The Hydrology of the UK (pp. 55-81). Routledge.

Serdar, C.C., Cihan, M., Yücel, D. and Serdar, M.A., 2021. Sample size, power and effect size revisited: simplified and practical approaches in pre-clinical, clinical and laboratory studies. Biochemia medica, 31(1), pp.27-53.

Shrivastava, S., Stewardson, M.J. and Arora, M., 2020. Distribution of clay-sized sediments in streambeds and influence of fine sediment clogging on hyporheic exchange. Hydrological Processes, 34(26), pp.5674-5685.

Shuker, L.J., Gurnell, A.M., Wharton, G., Gurnell, D. J., England, J., Finn Leeming, B.F. and Beach, E., 2017. MoRPh: a citizen science tool for monitoring and appraising physical habitat changes in rivers. Water and Environment Journal, 31(3), pp.418-424.

Sinclair, J.S., Mademann, J.A., Haubrock, P.J. and Haase, P., 2023. Primarily neutral effects of river restoration on macroinvertebrates, macrophytes, and fishes after a decade of monitoring. Restoration Ecology, 31(3), p.e13840.

Smith, B., Clifford, N.J. and Mant, J., 2014. The changing nature of river restoration. Wiley Interdisciplinary Reviews: Water, 1(3), pp.249-261.

Statzner, B., Bonada, N. and Dolédec, S., 2008. Conservation of taxonomic and biological trait diversity of European stream macroinvertebrate communities: a case for a collective public database. Biodiversity and conservation in Europe, pp.367-390.

Stubbs, A., 2021. British Craneflies. British Entomological and Natural History Society

Sykes, E.C., 1933. The Agricultural Geography of Northumberland. Geography, pp.269-282.

Szałkiewicz, E., Jusik, S. and Grygoruk, M., 2018. Status of and perspectives on river restoration in Europe: 310,000 Euros per hectare of restored river. Sustainability, 10(1), p.129.

Szałkiewicz, E., Kałuża, T. and Grygoruk, M., 2022. Detailed analysis of habitat suitability curves for macroinvertebrates and functional feeding groups. Scientific Reports, 12(1), p.10757.

Szarek-Gwiazda, E., Ciszewski, D. & Kownacki, A. 2023. 'The Effects of Channelization with Low In-Stream Barriers on Macroinvertebrate Communities of Mountain Rivers', Water (Basel), 15(6), p. 1059.

Tachet, H., Richoux, P., Bournaud, M. and Usseglio-Polatera, P., 2010. Invertébrés d'eau douce: systématique, biologie, écologie (Vol. 15, pp. 89-10). Paris: CNRS éditions.

Talbot, T. and Lapointe, M., 2002. Modes of response of a gravel bed river to meander straightening: The case of the Sainte-Marguerite River, Saguenay Region, Quebec, Canada. Water resources research, 38(6), pp.9-1.

Turner, S., Kumar, A., Smith, R., Bloomfield, J. and Clemas, S., 2024. Hydrological summary for the United Kingdom: May 2024.

Vermeulen, B., Hoitink, A.J.F. and Labeur, R.J., 2015. Flow structure caused by a local cross-sectional area increase and curvature in a sharp river bend. Journal of Geophysical Research: Earth Surface, 120(9), pp.1771-1783.

Waringer, J. & Graf, W. (2011) Atlas of Central European Trichoptera Larvae. Erik Mauch Verlag, Dinkelscherben

White, D.S., 2009. Coleoptera (Beetles) in aquatic ecosystems.

White, J.C., Hill, M.J., Bickerton, M.A. and Wood, P.J., 2017. Macroinvertebrate taxonomic and functional trait compositions within lotic habitats affected by river restoration practices. Environmental Management, 60, pp.513-525.

WHO., 2011. Guidelines for drinking-water quality. WHO chronicle, 38(4), pp.104-8.

Winiwarter, V., Schmid, M., Hohensinner, S. and Haidvogl, G., 2013. The environmental history of the Danube river basin as an issue of long-term socio-ecological research. Long term socio-ecological research: studies in society-nature interactions across spatial and temporal scales, pp.103-122.

Worrall, F., Burt, T.P. and Howden, N.J., 2016. The fluvial flux of particulate organic matter from the UK: the emission factor of soil erosion. Earth Surface Processes and Landforms, 41(1), pp.61-71.

Zhao, H.H., Shipp, A.J., Carter, K., Gonzalez-Mulé, E. and Xu, E., 2024. Time and change: A meta-analysis of temporal decisions in longitudinal studies. Journal of Organizational Behavior, 45(4), pp.620-640.

Zhou, T. and Endreny, T., 2020. The straightening of a river meander leads to extensive losses in flow complexity and ecosystem services. Water, 12(6), p.1680.

Appendix

List of Abbreviations

AECOM – Engineering/consultancy firm

ANOVA – Analysis of Variance

ASPT – Average Score Per Taxa

BMWP – Biological Monitoring Working Party Index

CCA – Canonical-correlation Analysis

EA – Environment Agency

EPT - Ephemeroptera (mayflies), Plecoptera (stoneflies), and Trichoptera (caddisflies)

EQI – Environment Quality Index

EQR – Environment Quality Ratios

EU LIFE – funding instrument by the European Union

LIFE – Lotic-Invertebrate Index for Flow Evaluation

LIFE WADER – EU LIFE project (Water and Disturbance Environment Restoration on Northumberland coast)

MINTA – Minium of Ntaxa and ASPT

MoRPh – Modular River Habitat Survey

nMDS – Non-Metric Multidimensional Scaling

Ntaxa – Number of taxa

ORP – Oxidation-Reduction Potential

PCA – Principal Component Analysis

PERMANOVA – Permutational analysis of variance

PSI – Proportion of Sediment-sensitive Invertebrates

RICT – River Invertebrate Classification Tool

RIVPACS – River Invertebrate Prediction & Classification Systems

SAC – Special Areas of Conservation

SIMPER – Similarity Percentage

SSSI – Site of Special Scientific Interest

TDS – Total Dissolved Solids

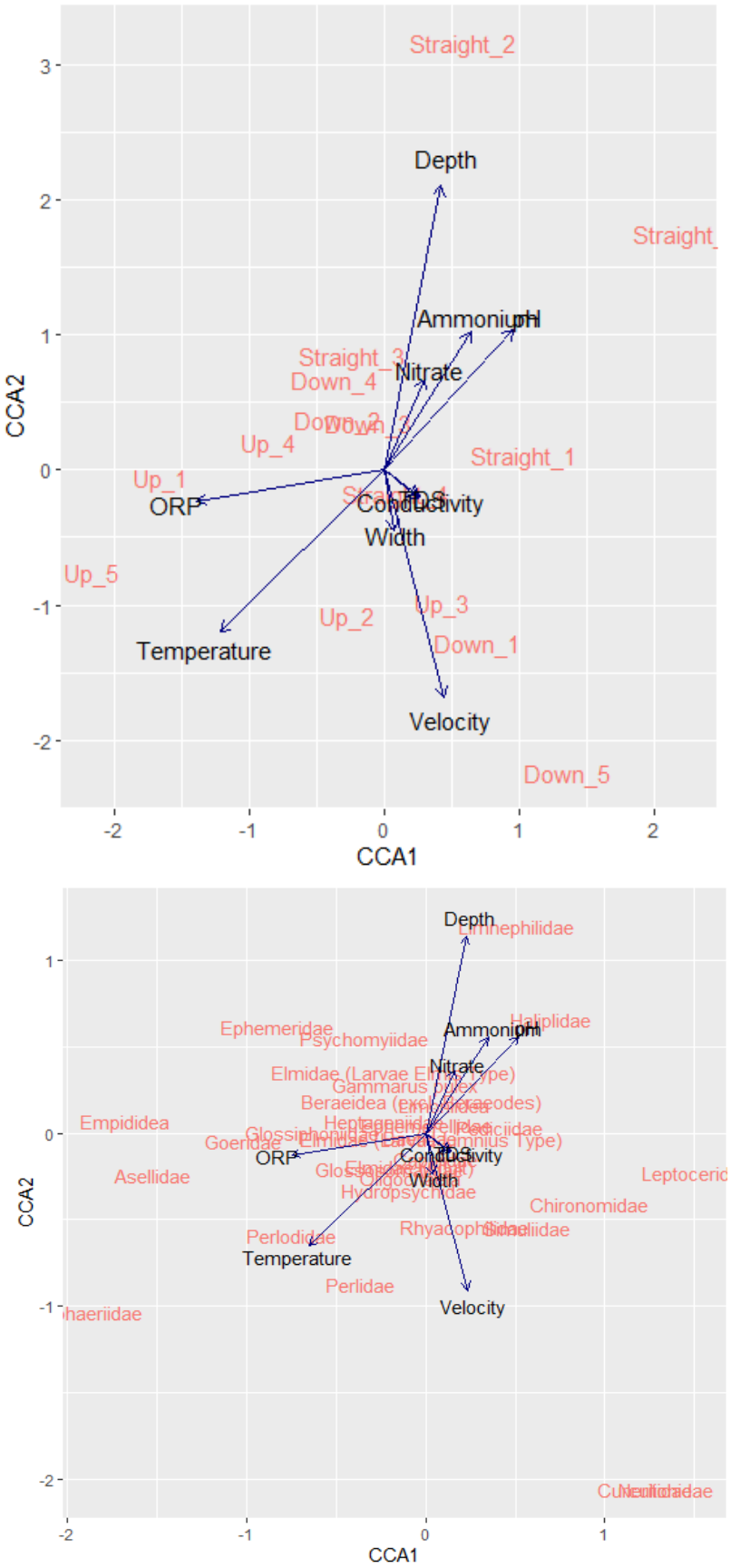
WHPT – Walley Hawkes Paisley Trigg Index

Supplementary Material

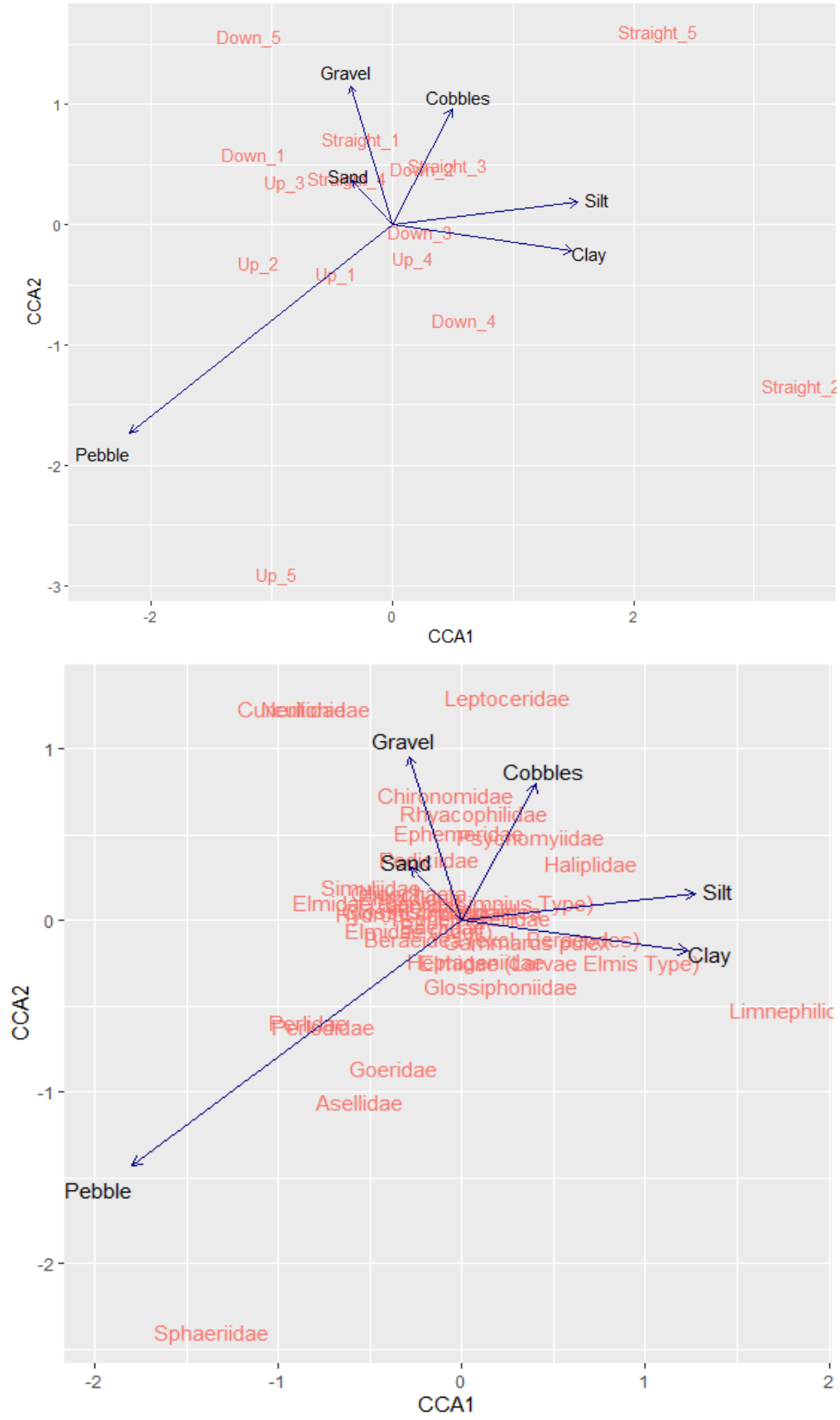
Appendix 1. Cumulative full output of SIMPER analysis between the abundance of macroinvertebrate taxa across the three different sample areas. Highlighted values indicate the largest values in comparison. * $p < 0.05$. ** $p < 0.001$.

Upstream vs Straightened				Straightened vs Downstream				Upstream vs Downstream			
Species	ava	avb	Cu. Sum	Species	ava	avb	Cu. Sum	Species	ava	avb	Cu. Sum
Baetidae	0.4945	0.4325	10.8%	Baetidae	0.4325	0.4209	10.9%	Simuliidae	0.0908	0.1086	10.3%
Elmidae (Larvae Limnius)	0.1736	0.1475	17.6%	Elmidae (Larvae Limnius)	0.1475	0.2625	20.3% *	Elmidae (Larvae Limnius)	0.1736	0.2625	19.2%
Ephemerellidae	0.7091	0.7123	23.7%	Simuliidae	0.0413	0.1086	27.7%	Baetidae	0.4945	0.4209	26.2%
Elmidae (Adult)	0.1625	0.0775	29.4%	Elmidae (Adult)	0.7750	0.1704	34.2% .	Ephemerellidae	0.7091	0.7523	31.8%
Haliplidae	0	0.0897	35% **	Ephemerellidae	0.7123	0.7523	40.5%	Heptageniidae	0.1746	0.1409	37.4%
Simuliidae	0.0908	0.0413	40.5%	Beraeidea (excl. Beraeodes)	0.1888	0.1032	46.1% .	Hydropsychidae	0.1441	0.0903	42.8%
Beraeidea (excl. Beraeodes)	0.1217	0.1888	45.9% .	Haliplidae	0.0897	0.0154	51.6% *	Capniidae	0.1307	0.0753	47.8%
Hydropsychidae	0.1441	0.0638	50.9% *	Gammarus pulex	0.1192	0.0934	56.4%	Beraeidea (excl. Beraeodes)	0.1217	0.1032	52.3%
Limoniidea	0.1851	0.2666	55.9% **	Limoniidea	0.2666	0.1957	61.1% *	Perlodidae	0.0548	0.0222	56.5% .
Heptageniidae	0.1746	0.1242	60.4%	Hydropsychidae	0.0638	0.0903	65.3%	Elmidae (Adult)	0.1625	0.1704	60.6%
Gammarus pulex	0.1038	0.1192	64.6%	Capniidae	0.1052	0.0753	69.5%	Elmidae (Larvae Elmis)	0.0494	0.0389	64%
Pediciidae	0.0211	0.0664	68.4%	Pediciidae	0.0664	0.0491	73.4%	Oligochaeta	0.0405	0.0150	67.3%
Limnephilidae	0	0.0552	71.9%	Limnephilidae	0.0552	0.0089	77.2% .	Pediciidae	0.0211	0.0491	70.6%
Perlodidae	0.0548	0	75.3% *	Heptageniidae	0.1242	0.1409	80.6%	Gammarus pulex	0.1038	0.0934	73.7%
Capniidae	0.1307	0.1052	78.6%	Glossosmatidae	0.0323	0.0439	83.4%	Chronomidae	0	0.0381	76.7%
Glossosmatidae	0.0632	0.0323	81.6% *	Chronomidae	0.0272	0.0381	86%	Limoniidea	0.1851	0.1957	79.7%
Elmidae (Larvae Elmis)	0.4940	0.0448	84.6%	Elmidae (Larvae Elmis)	0.0448	0.0389	88.4%	Perlidae	0.0351	0.0181	82.3%
Oligochaeta	0.0405	0.0165	87.1%	Oligochaeta	0.0165	0.0150	90%	Psychomyiidae	0.0252	0.0150	84.9%
Perlidae	0.0351	0	89.2%	Psychomyiidae	0.0148	0.0150	91.6%	Glossosmatidae	0.0632	0.0439	87.2%

Psychomyiidae	0.0253	0.0148	91.1% *	Perlodidae	0	0.0222	93%	Goeridae	0.0258	0	89.3%
Goeridae	0.0258	0.0085	92.8%	Leptoceridae	0.1870	0	94.3%	Asellidae	0.0204	0.0089	91.1%
Chronomidae	0	0.0272	94.4% *	Perlidae	0	0.0181	95.5%	Rhyacophilidae	0.0095	0.0167	92.7%
Asellidae	0.0204	0	95.7%	Rhyacophilidae	0	0.0167	96.5%	Haliplidae	0	0.0154	94%
Leptoceridae	0	0.0187	96.9%	Glossiphoniidae	0	0.0102	97.2% .	Ephemeridae	0.0125	0	95%
Ephemeridae	0.0125	0.0085	97.9%	Asellidae	0	0.0089	97.8%	Empididea	0.0125	0	96.1%
Empididea	0.0125	0	98.7% .	Curculionidae	0	0.0092	98.4% .	Sphaeriidae	0.0115	0	97%
Sphaeriidae	0.0115	0	99.4% .	Neritidae	0	0.0092	99% .	Glossiphoniidae	0	0.0102	97.8%
Rhyacophilidae	0.0095	0	100%	Ephemeridae	0	0	99.5%	Limnephilidae	0	0.0089	98.6%
Curculionidae	0	0	100%	Goeridae	0	0	100%	Curculionidae	0	0.0092	99.3%
Neritidae	0	0	100%	Empididea	0	0	100%	Neritidae	0	0.0092	100%
Glossiphoniidae	0	0	100%	Sphaeriidae	0	0	100%	Leptoceridae	0	0	100%



Appendix 2. CCA graphs of macroinvertebrate abundance related to chemical and hydromorphic variables (not including substrate). (top) By Site. (bottom) By Species.

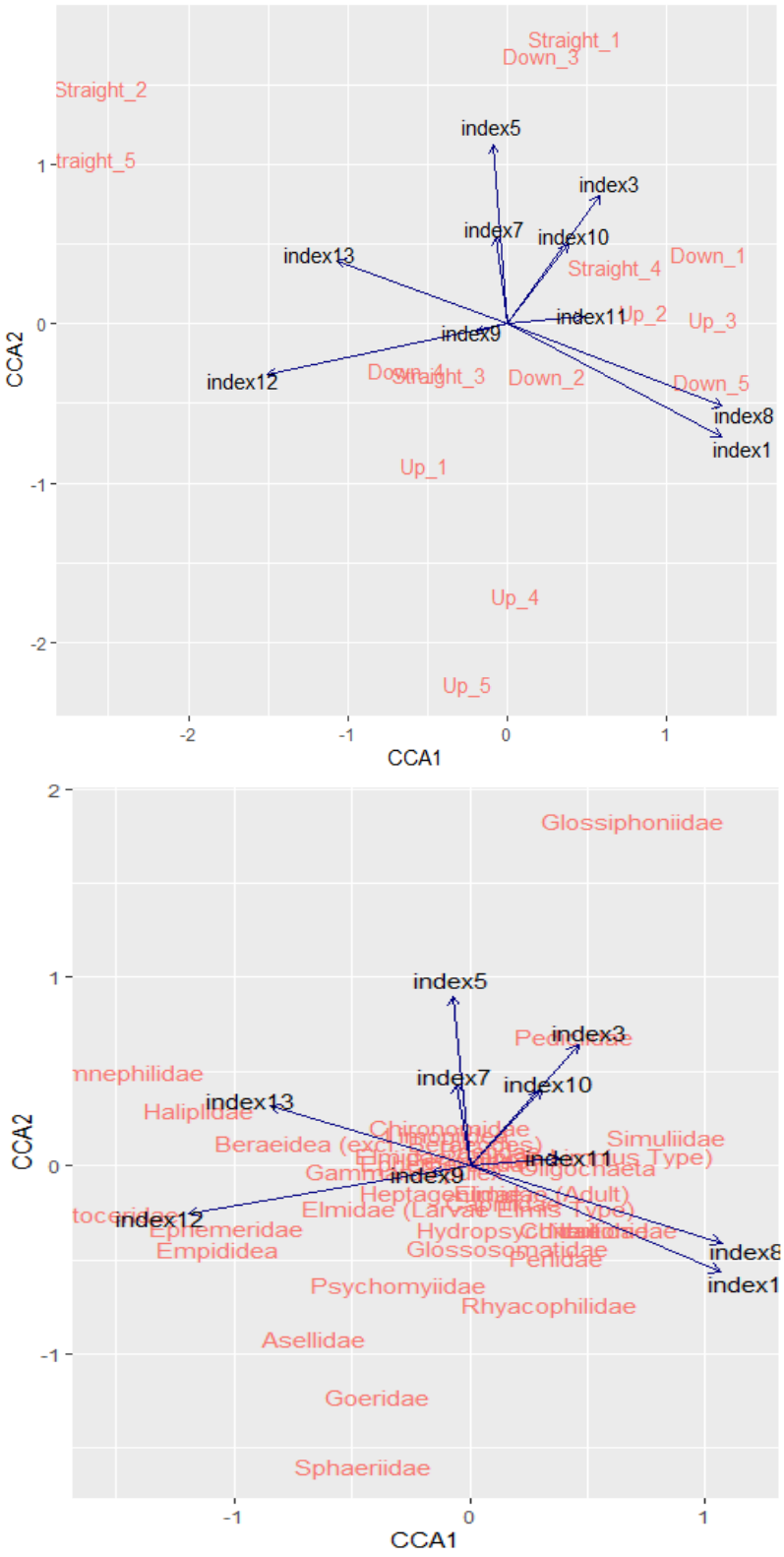


Appendix 3. CCA graphs of the macroinvertebrate abundances with substrate composition. (top) By Site. (bottom) By Species.

Appendix 4. Cumulative full output of SIMPER analysis of macroinvertebrate functional traits between the three sample areas. Highlighted values indicate the largest values in comparison. * $p < 0.05$. ** $p < 0.001$. *** $p < 0.0001$.

Upstream vs Straightened				Straightened vs Downstream				Upstream vs Downstream			
Trait	ava	avb	Cu. Sum	Trait	ava	avb	Cu. Sum	Trait	ava	avb	Cu. Sum
RF5	1.4082	1.3904	4.7% .	RF5	1.3904	1.3666	4.4%	MPS3	1.6008	1.4937	4.1% *
LCD2	0.5925	0.4864	8.2%	LCD1	2.0846	2.0190	8.2% .	LCD1	2.0994	2.0190	8.1%
FH4	0.9399	0.8618	11.5% *	FH4	0.8618	0.9549	11.6% *	FH5	0.2261	0.1798	11.8% .
RC2	1.7473	1.7164	14.9%	RC2	1.7164	1.7130	14.8%	FH3	0.8615	0.7557	15.5% *
R4	1.0972	1.0567	18%	FH5	0.1183	0.1798	17.7%	LSR7	0.2063	0.1698	18.8%
LSR4	1.7035	1.6115	21.2%	MPS3	1.5135	1.4937	20.6%	LSR4	1.7035	1.6045	21.9%
R3	0.9147	0.8074	24.2% ***	FH3	0.8486	0.7557	23.5%	R4	1.0972	1.0273	25.1%
CV4	0.6965	0.5987	27% *	R4	1.0567	1.0273	26.3%	MPS2	0.4015	0.4730	28%
FH5	0.2261	0.1183	29.9%	LSR7	0.1193	0.1698	29%	RF5	1.4082	1.3666	30.9%
RC1	0.2120	0.1105	32.6% **	LCD2	0.4864	0.5215	31.7%	RF4	0.4039	0.3704	33.7%
MPS3	1.6008	1.5135	35.1%	LSR4	1.6115	1.6045	34.4%	MPS4	0.5411	0.4476	36.5%
F4	0.7901	0.7459	37.6%	RS4	0.3363	0.3704	37%	F2	0.4127	0.3864	39.2%
MPS2	0.4015	0.4271	40.1%	F2	0.3283	0.3864	39.5%	RC2	1.7473	1.7130	41.7%
CV3	0.8588	0.7791	42.5% .	CV4	0.5987	0.6520	42%	RF1	0.8618	0.8009	44.3%
LCD1	2.0994	2.0846	44.9%	F4	0.7459	0.7849	44.3%	LCD2	0.5925	0.5315	46.8%
AS1	1.2038	1.1464	47.2% .	MPS2	0.4271	0.4730	46.4%	RC1	0.2120	0.1332	49.1%
F2	0.4127	0.3283	49.6%	CV3	0.7791	0.8049	48.4%	R1	0.2341	0.1637	51.4%
LSR7	0.2063	0.1193	51.9%	R1	0.2016	0.1637	50.4%	AS2	1.2152	1.1487	53.7%
MPS4	0.5411	0.4709	54.2%	LSR5	0.2542	0.1841	52.4% *	AS1	1.2038	1.1288	55.9%
S1	0.5774	0.4997	56.5% *	AS2	1.1593	1.1487	54.3%	AS3 *	0.1643	0.1499	58% .
AS2	1.2152	1.1593	58.7%	RC3	0.7441	0.7042	56.1%	R3	0.9147	0.8726	60%
RF1	0.8618	0.8351	60.8%	RF1	0.8351	0.8009	58%	CV4	0.6965	0.6520	62%
LSR5	0.1775	0.2542	62.9% *	AC1	1.1464	1.1288	59.8%	RC3	0.7326	0.7042	64%
RF4	0.4039	0.3363	64.8%	AC3	0.1546	0.1499	61.7% .	S1	0.5774	0.5308	65.9%
R1	0.2341	0.2016	66.7%	R3	0.8074	0.8726	63.5%	CV3	0.8588	0.8049	67.7%
FH3	0.8615	0.8486	68.6%	CV1	0.3434	0.2805	65.3% *	D2	0.7336	0.6877	69.3%
CV1	0.2975	0.3434	70%	S1	0.4997	0.5308	67%	D3 *	0.3553	0.3459	70.9% .
F3	0.4701	0.4781	71.4%	F3	0.4781	0.4297	68.8% .	F3	0.4701	0.4297	72.3%
CV2	0.8391	0.8499	72.8%	MPS4	0.4709	0.4476	70.3%	R5	0.0581	0.0716	73.8%
D2	0.7336	0.6927	74.2%	D3	0.3646	0.3459	71.7% *	FH7	0.2614	0.2717	75.3%
S2	0.3585	0.3208	75.6% *	CV2	0.8499	0.8130	73.1%	D4	0.8096	0.7650	76.7%
D4	0.8096	0.7890	76.9%	D2	0.6927	0.6877	74.5%	F4	0.7901	0.7849	78.1%
RC3	0.7326	0.7441	78.2%	FH7	0.2764	0.2717	75.9%	FH4	0.9399	0.9549	79.5%

F9	0.1376	0.1202	79.5%	RC1	0.1105	0.1332	77.2%	S5	0.5576	0.5418	80.8%
FH7	0.2614	0.2764	80.7%	R5	0.0550	0.0716	78.5%	CV1	0.2975	0.2805	82.1%
LSR6	0.1879	0.1493	81.9% .	D4	0.7890	0.7650	79.8%	F8	0.1489	0.1596	83.3%
F8	0.1489	0.1745	83%	F5	0.4669	0.4204	81.1% *	F7	0.1739	0.1581	84.5%
R2	0.3725	0.4091	84.1% .	F9	0.1202	0.1170	82.3%	MPS5	0.1486	0.1362	85.6%
S3	0.3155	0.3010	85.2% .	FH2	0.4116	0.3703	83.5% **	F5	0.4552	0.4204	86.7%
MPS5	0.1486	0.1595	86.3%	S2	0.3208	0.3324	84.7%	LSR5	0.1775	0.1841	87.8%
S5	0.5576	0.5222	87.3%	MPS5	0.1595	0.1362	85.8%	CV2	0.8391	0.8130	88.8%
FH6	0.0105	0.0502	88.4% *	S5	0.5222	0.5418	86.9%	R2	0.3725	0.3830	89.7%
R5	0.0581	0.0550	89.4%	FH6	0.0502	0.0150	87.9% *	S7	0.3515	0.3338	90.5%
F7	0.1739	0.1377	90.4%	S8	0.2854	0.2474	88.9% *	S2	0.3585	0.3324	91.4%
R7	0.0152	0.0413	91.3%	LSR6	0.1493	0.1778	89.9%	FH2	0.3925	0.3704	92.2%
FH2	0.3925	0.4116	92.1%	S3	0.3010	0.2913	91%	R7	0.0152	0.0321	92.9%
AS4	0.1088	0.1108	93%	F8	0.1745	0.1596	92%	S3	0.3155	0.2913	93.7%
AS3	0.1643	0.1546	93.8%	F7	0.1377	0.1581	92.9%	LSR6	0.1879	0.1778	94.5%
S8	0.2607	0.2854	94.5%	R7	0.0413	0.0321	93.8%	AS4	0.1088	0.1231	95.2%
LSR3	0.3670	0.3880	95.2%	LSR3	0.3880	0.3558	94.7% *	F9	0.1376	0.1170	95.9%
F5	0.4552	0.4669	95.8%	R2	0.4091	0.3830	95.5%	S8	0.2607	0.2474	96.6%
F6	0.0703	0.0877	96.4% *	AS4	0.1108	0.1231	96.4%	LSR3	0.3670	0.3558	97.3%
S9	0.1077	0.1304	97% **	S7	0.3356	0.3338	97.1%	F6	0.0703	0.0641	97.6%
S4	0.1363	0.1566	97.5%	F6	0.0877	0.0641	97.7%	LSR1	0.0423	0.0559	98.1%
S7	0.3515	0.3356	98%	S9	0.1304	0.1112	98.2%	S9	0.1077	0.1112	98.4%
D3	0.3553	0.3646	98.5%	S4	0.1566	0.1407	98.7%	FH6	0.0105	0.0150	98.7%
RS2	0.0181	0.0092	98.9% *	LSR1	0.0452	0.0559	99.1%	S4	0.1363	0.1407	99%
LSR1	0.0423	0.0452	99.2%	RF2	0.0092	0.0126	99.4%	S6	0.0268	0.0211	99.3%
S6	0.0268	0.0193	99.6%	S6	0.0193	0.0211	99.6%	RF2	0.0181	0.0126	99.5%
LSR8	0.0075	0.0035	99.8%	F1	0.0316	0.0301	99.7%	LSR8	0.0075	0.0025	99.8%
F1	0.0331	0.0316	99.9%	FH8	0.0042	0.0029	99.9%	F1	0.0331	0.0301	99.9%
FH8	0	0.0042	100%	LSR8	0.0035	0.0025	100%	FH8	0	0.0029	100%



Appendix 5. CCA graphs of the macroinvertebrate abundances with the MoRPh indices. (top) By Site. (bottom) By Species.